VIETNAM GENERAL CONFEDERATION OF LABOUR
TON DUC THANG UNIVERSITY
FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING



DO HOANG MINH HAN

DESIGN AND IMPLEMENTATION OF MOTORCYCLE POSITIONING AND ANTI-THEFT DEVICE USING GPS AND GSM TECHNOLOGY

EMBEDDED SYSTEM PROJECT ELECTRONICS AND TELECOMMUNICATIONS ENGINEERING

HO CHI MINH CITY, 2026

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Advisor:
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HO CHI MINH CITY, 2026

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Ho Chi Minh City, date 6 month 5 year 2026

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DESIGN AND IMPLEMENTATION OF MOTORCYCLE POSITIONING AND ANTI-THEFT DEVICE USING GPS AND GSM TECHNOLOGY

ABSTRACT

Motorcycle theft remains a practical security problem, especially when traditional protection methods are not sufficient to provide timely warning and location monitoring. This project presents the design and implementation of a motorcycle positioning and anti-theft device using Global Positioning System (GPS) and Global System for Mobile Communications (GSM) technology, with the Peripheral Interface Controller 16F877A (PIC16F877A) microcontroller serving as the central control unit. The proposed system is capable of receiving GPS coordinate data, extracting latitude and longitude information, and displaying the current position on a Liquid Crystal Display (LCD). To improve reliability in practical operating conditions, the project also applies a geofencing function to detect when the motorcycle moves outside a predefined safe zone, as well as a last known position mechanism to preserve the most recent valid coordinates when GPS signal loss occurs. In addition, a vibration sensor is used to detect suspicious physical impact or tampering. When an abnormal event is detected, such as unusual vibration or unauthorized movement beyond the safe zone, the system activates a local buzzer alarm and sends a Short Message Service (SMS) warning message containing the relevant position information to the owner's phone. The hardware structure, software control flow, and operating principle of the system are designed and implemented in an experimental model. The results of this project show that the proposed embedded system can provide a simple, practical, and effective solution for motorcycle monitoring and anti-theft warning under basic real-world conditions.

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LIST OF ABBREVIATIONS

ACC	Accessory
ADC	Analog-to-Digital Converter
AT	Attention
CPU	Central Processing Unit
DC	Direct Current
DIP	Dual In-line Package
EEPROM	Electrically Erasable Programmable Read-Only Memory
GPIO	General-Purpose Input/Output
GSM	Global System for Mobile Communications
GPS	Global Positioning System
I/O	Input/Output
I2C	Inter-Integrated Circuit
IoT	Internet of Things
LCD	Liquid Crystal Display
LED	Light-Emitting Diode
LTE	Long-Term Evolution
MCLR	Master Clear Reset
MCU	Microcontroller Unit
NMEA	National Marine Electronics Association
PCB	Printed Circuit Board
PIC	Peripheral Interface Controller
PWM	Pulse Width Modulation
RAM	Random Access Memory
SMS	Short Message Service
SPI	Serial Peripheral Interface
TTL	Transistor-Transistor Logic
UART	Universal Asynchronous Receiver and Transmitter
UBX	u-blox Binary Protocol
USART	Universal Synchronous and Asynchronous Receiver and Transmitter
USB	Universal Serial Bus

CHAPTER 1. INTRODUCTION

1.1 Rationale for Choosing the Topic

Motorcycles are one of the most common means of transportation in Vietnam, so protecting personal vehicles is an important practical concern. In reality, motorcycle theft still occurs frequently, while theft methods are becoming more sophisticated. Because of this, traditional protection methods such as mechanical locks are often not sufficient. With the development of embedded systems and wireless communication technologies, combining Global Positioning System (GPS) and Global System for Mobile Communications (GSM) in a compact anti-theft device provides a practical solution for motorcycle monitoring and warning. For this reason, the topic "Design and Implementation of a Motorcycle Positioning and Anti-Theft Device Using GPS and GSM Technology" was selected. Besides its practical value, this project also provides a good opportunity to apply embedded-system knowledge to a real engineering problem.

Another reason for choosing this topic is that it supports the development of professional skills in both hardware and software. Through this project, the author can study the operation of the PIC microcontroller, interface it with peripheral modules, process GPS data, and implement Short Message Service (SMS)-based warning functions. At the same time, the project helps improve system design thinking, problem-solving ability, and practical experience in circuit implementation, programming, testing, and debugging.

1.2 Project Objectives

The main objective of this project is to study, design, and implement a motorcycle monitoring and anti-theft device that can operate stably under practical condit-ions. In terms of hardware, the project aims to design the circuit schematic and implement a complete Printed Circuit Board (PCB) for the main functional blocks, including the PIC16-F877A microcontroller, GPS module, GSM module, vibration sensor, and buzzer alarm. The hardware must be connected properly, arranged in a compact form, and supplied with suitable and stable operating voltages. In terms of software, the project focuses on developing a control program in C language for system manag-ement. The program must be able to read

and process GPS data, detect abnormal vibration, store the current or last valid position, and send an SMS warning message containing location information when an intrusion or unauthorized movement is detected.

1.3 Scope of the Project

This project focuses on the design and implementation of a motorcycle positioning and anti-theft device using GPS and GSM technology. The proposed system is built around the PIC16F877A microcontroller and includes the main modules required for positioning, warning, and communication. In terms of hardware, the project is limited to circuit design, PCB implementation, and testing of the main modules in an experimental model. In terms of software, the work is limited to developing a control program for GPS data reading, abnormal vibration detection, position storage, and SMS warning generation.

The project does not include the development of a mobile application or a web-based monitoring platform. The main goal is to create a simple, practical, and stable embedded model that can support basic motorcycle monitoring and anti-theft warning functions.

1.4 Idea and Implementation Method

The project integrates positioning and anti-theft warning functions into a single embedded system. In addition to location tracking, the proposed model can detect abnormal vibration or unauthorized movement and generate a warning when needed. The system includes the PIC16F877A microcontroller, GPS module, GSM module, vibration sensor, and buzzer alarm. In this design, the microcontroller acts as the central processing unit, while the GPS module provides location data, the vibration sensor detects suspicious impacts, and the GSM module sends warning messages to the user.

The implementation covers both hardware and software development. On the hardware side, the project includes circuit design, module interfacing, and PCB implementation. On the software side, a C program is developed to process GPS data, detect abnormal events, store valid position information, and generate SMS warnings. After completion, the system is assembled and tested under simulated conditions to evaluate its performance and identify possible improvements.

CHAPTER 2. THEORETICAL BACKGROUND AND SYSTEM MODEL

2.1 Overview of Motorcycle Tracking and Anti-Theft Systems

A motorcycle tracking and anti-theft system is an embedded model that combines positioning, abnormal-event detection, and user warning functions. In this project, the system is designed to monitor the current location of the motorcycle and detect suspicious situations such as strong vibration, tampering, or movement outside a predefined safe zone.

The operating principle is based on a microcontroller that receives and processes data from the main functional blocks. The GPS module provides latitude and longitude data, the vibration sensor detects unusual physical impacts, and the GSM module is used to send SMS warnings to the owner. In addition, a buzzer is used for local alarm signaling when an abnormal condition is detected.

Because GPS reception may become weak or unavailable in places such as basements, covered parking areas, or indoor environments, the system also includes a last known position mechanism. This allows the device to preserve the most recent valid coordinates and maintain basic monitoring capability when new GPS data are not available.

In general, the proposed system performs five main functions: receiving GPS coordinates, displaying basic information, detecting abnormal vibration, storing the last valid position, and sending warning messages when an abnormal event occurs. These functions form the basis for the component analysis and operating principles presented in the following sections.

2.2 Main Technologies Used in the Project

To develop this system, the project uses three main technologies: embedded systems, GPS positioning technology, and GSM communication through SMS messages. Each technology has its own role, but when combined, they form a complete system that can monitor location, detect abnormal conditions, proactiv-ely identify possible intrusion risks, and send timely warnings to the user.

2.2.1. Geofencing Principle and Safe-Zone Distance Calculation

To improve the anti-theft capability of the proposed system, a geofencing method is applied to determine whether the motorcycle remains inside a predefined safe zone. In this project, the safe zone is defined as a circular region centered at the reference parking position of the motorcycle. When the measured position of the vehicle exceeds the allowed radius, the system considers that the motorcycle has been moved abnormally and triggers a warning process. A similar geofencing concept has also been applied in location-based attendance systems, where the user position is compared with a predefined geographic boundary Rao et al. (2022).

Let the reference safe position be denoted by

$$(Lat_0, Lon_0) \tag{2.1}$$

where Lat_0 and Lon_0 are the latitude and longitude recorded when the motorcycle is parked and the anti-theft mode is activated. During operation, the GPS module continuously provides the current position of the motorcycle, denoted by

$$(Lat_1, Lon_1) \tag{2.2}$$

where Lat_1 and Lon_1 are the current latitude and longitude.

According to the National Marine Electronics Association (NMEA) standard, latitude and longitude are output in the format of degrees, minutes, and decimal fractions of minutes. Therefore, before performing any distance calculation, the received coordinates must be converted into decimal degrees u-blox AG (2023) .

$$Lat = D_{Lat} + \frac{M_{Lat}}{60} \tag{2.3}$$

$$Lon = D_{Lon} + \frac{M_{Lon}}{60} \tag{2.4}$$

where D_{Lat} and D_{Lon} are the degree parts, while M_{Lat} and M_{Lon} are the minute parts of the latitude and longitude, respectively.

After conversion to decimal degrees, the coordinate values are transformed from degrees to radians before applying the Haversine formula Naik and Nair (2019). Thus,

$$\phi_0 = Lat_0 \cdot \frac{\pi}{180}, \qquad \lambda_0 = Lon_0 \cdot \frac{\pi}{180} \tag{2.5}$$

$$\phi_1 = Lat_1 \cdot \frac{\pi}{180}, \qquad \lambda_1 = Lon_1 \cdot \frac{\pi}{180} \tag{2.6}$$

The latitude and longitude differences are then given by

$$\Delta\phi = \phi_1 - \phi_0 \tag{2.7}$$

$$\Delta\lambda = \lambda_1 - \lambda_0 \tag{2.8}$$

To calculate the actual distance between the reference position and the current position on the Earth's surface, the Haversine formula is used Naik and Nair (2019). The Haversine coefficient is written as

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_0)\cos(\phi_1)\sin^2\left(\frac{\Delta\lambda}{2}\right) \tag{2.9}$$

The great-circle distance between the two coordinates is then obtained as

$$d = 2R\arcsin(\sqrt{a}) \tag{2.10}$$

where d is the actual distance between the safe position and the current position, and R is the Earth's radius. In practical Haversine-based computation, the radius of the Earth is commonly approximated by

$$R = 6,371,000 \text{ m} \tag{2.11}$$

which corresponds to 6371 km in the implementation reported by Naik and Nair Naik and Nair (2019).

In this project, the safe-zone radius is selected as

$$d_{safe} = 10 \text{ m} \tag{2.12}$$

Accordingly, the decision rule of the system is defined as follows: If $d \leq d_{safe}$, the motorcycle is considered to remain inside the safe zone. If $d > d_{safe}$, the motorcycle is considered to have moved outside the safe zone.

This distance-versus-radius comparison is consistent with geofencing-based decision logic, where the measured distance is checked against a predefined geofence radius to determine whether the object is inside or outside the allowed area Rao et al. (2022).

When the condition $d > 10$ m is detected, the microcontroller activates the buzzer alarm and simultaneously sends an SMS warning message to the owner

through the GSM module. The warning message may include the current coordinates of the motorcycle so that the owner can identify its location quickly.

The Haversine formula is selected in this project because it provides a practical and well-established way to estimate the real displacement between two GPS coordinates on the Earth's surface Naik and Nair (2019). In addition, Naik and Nair reported that the uBlox NEO-6M receiver used in their study has a maximum navigation update rate of 5 Hz and a horizontal position accuracy of about 2.5 m, which confirms that GPS readings may fluctuate in practical measurements and therefore should be interpreted carefully when the threshold is small Naik and Nair (2019). For this reason, the out-of-safe-zone condition may additionally be verified over several consecutive valid samples in practical implementation in order to reduce false alarms caused by short-term GPS fluc-tuation.

2.2.2. Last Known Position Handling During GPS Signal Loss

In practical use, GPS signals may become weak or temporarily unavailable when the motorcycle is parked indoors, in a basement, or near surrounding obstacles. In such cases, the GPS receiver may not continuously provide valid coordinates. To maintain the monitoring function of the system, a last known position mechanism is applied.

Let the latest valid position be denoted by

$$(Lat_{last}, Lon_{last})$$

where Lat_{last} and Lon_{last} are the latest valid latitude and longitude obtained from the GPS receiver. Whenever valid NMEA data are available, the microcontroller updates this stored position. If the GPS signal is lost, the system does not overwrite the stored data with invalid values but keeps the last valid position as the most reliable reference Naik and Nair (2019).

This mechanism improves practical reliability in two ways. First, it prevents the Liquid Crystal Display (LCD) from displaying empty or incorrect coordinates when a valid GPS fix is unavailable. Second, if an abnormal event occurs during signal loss, the system can still send an SMS warning containing the most recent valid location of the motorcycle.

In operation, valid GPS data are used to update the current position, store it in memory, and calculate the distance from the safe reference point. If the

calculated distance exceeds the safe-zone threshold, the buzzer alarm and SMS warning are activated. Otherwise, when valid GPS data are not available, the system continues using the last stored position for display and warning support. By combining geofencing with last known position storage, the proposed system can operate more reliably under real conditions and provide more practical anti-theft support Naik and Nair (2019).

2.2.3. Embedded System Concept

An embedded system is an electronic system designed to perform specific functions within a dedicated device. Unlike a general-purpose computer, it is developed for a limited set of tasks and is expected to operate continuously and reliably.

In this project, the embedded system is built around the PIC16F877A microcontroller, which acts as the central processing unit. It receives data from the GPS module, vibration sensor, and GSM module, processes the information, and controls the corresponding responses. This approach makes the proposed model compact, cost-effective, and suitable for practical motorcycle monitoring and anti-theft applications.

2.2.4. GPS Positioning Technology

GPS is a positioning technology that determines the location of an object from satellite signals. The GPS module calculates the current position in the form of latitude and longitude and transmits this information to the microcontroller for further processing.

In this project, GPS data are used for coordinate display, location monitoring, and safe-zone detection. Because signal reception may become weak or unstable in some environments, the system also applies a last known position mechanism to maintain basic operation when valid GPS data are temporarily unavailable.

2.2.5. GSM-Based SMS Communication

GSM technology is used in this project to support remote warning transmission through SMS messages. When an abnormal event such as unusual vibration or unauthorized movement is detected, the microcontroller sends AT commands to the communication module so that a warning message can be delivered to the owner. The message may contain the current coordinates or the last known

valid position of the motorcycle. Together with the embedded control and GPS positioning functions, this communication function completes the proposed anti-theft system.

2.3 PIC16F877A Microcontroller

PIC16F877A is an 8-bit microcontroller from the PIC family developed by Microchip. It is widely used in embedded applications because of its stable performance, simple programming process, and the integration of many useful peripheral functions. In this project, the PIC16F877A microcontroller is selected as the central processing unit, responsible for collecting data from external modules, processing information, and controlling the overall operation of the motorcycle tracking and anti-theft system.



Figure 2.1. Hardware of the PIC16F877A microcontroller

The actual hardware of the PIC16F877A microcontroller is shown in Figure 2.1. This device is provided in a 40-pin package, which offers enough connection pins for power supply, oscillator, reset, and input/output functions. With this pin configuration, the microcontroller can interface with different external modules used in the project, such as the GPS module, GSM module, LCD, vibration sensor, and alarm buzzer. This makes PIC16F877A suitable for the hardware connection requirements of the proposed system.

In terms of internal structure, PIC16F877A includes important functional

blocks such as program memory, Random Access Memory (RAM), Electrically Erasable Programmable Read-Only Memory (EEPROM), timers, an Analog-to-Digital Converter (ADC), an interrupt system, and serial communication interfaces. Thanks to these built-in features, the microcontroller can perform multiple tasks within the same system, such as reading sensor data, processing location information from the GPS module, controlling the LCD display, and sending commands to the GSM module. The main technical specifications of this micro-controller are presented in Table 2.1.

Table 2.1. PIC16F877A Parameters

Parameter	**Value**
CPU Architecture	8-bit PIC microcontroller
Program Memory	8K × 14 words Flash
Data RAM	368 bytes
Data EEPROM	256 bytes
Maximum Clock Frequency	Up to 20 MHz
Operating Voltage	2.0V – 5.5V DC
I/O Pins	33 digital I/O pins
ADC	8-channel, 10-bit ADC
Timers	3 timers (TMR0, TMR1, TMR2)
Serial Communication	1 USART, SPI, I2C
Package	40-pin DIP
Operating Temperature	-40°C to +85°C

The main technical specifications of the PIC16F877A microcontroller are summarized in Table 2.1, adapted from the Microchip PIC16F87XA Data Sheet Microchip Technology Inc. (2003).

In addition, PIC16F877A supports an interrupt mechanism, which helps the system respond more quickly when an event occurs. This microcontroller also provides one hardware Universal Asynchronous Receiver and Transmitter (UART) for serial communication with peripheral devices. In the final practical implementation of this project, the single hardware UART is reused for two serial tasks

at different stages of operation. The GPS module sends coordinate data to the UART receive line of the microcontroller, while the UART transmit line of the microcontroller is used to send command strings to the mobile communication module for SMS warning transmission. This arrangement avoids the instability of software UART and provides a more reliable solution for the experimental model.

2.4 NEO-6M GPS Module and Positioning Principle

The project uses the NEO-6M GPS module with an external antenna Figure 2.2 as the positioning unit of the system. This version is selected because the external antenna makes installation more flexible in the practical model and helps the module receive satellite signals more conveniently in real use. According to the NEO-6 documentation, the NEO-6 family supports both passive and active antennas, which makes it suitable for different installation conditions.

In terms of communication, this module transmits data through UART serial communication. According to the NEO-6 documentation, the module supports standard NMEA output and a configurable UART interface, while common GY-NEO6MV2 boards are typically used at a default baud rate of 9600 bps. From these data strings, the microcontroller extracts the coordinate information required for monitoring and warning functions. The main technical specifications of the module are presented in Table 2.2.

The main specifications of the NEO-6M GPS receiver used in this project are summarized in Table 2.2, based on the official u-blox NEO-6 data sheet u-blox AG (2011).

However, the GPS signal may become weak or temporarily unavailable when the motorcycle is indoors, in a basement, or in a covered area. To overcome this limitation, the system is equipped with a mechanism to store the last valid position. When no new satellite data are available, the microcontroller uses the stored position to maintain basic monitoring capability.

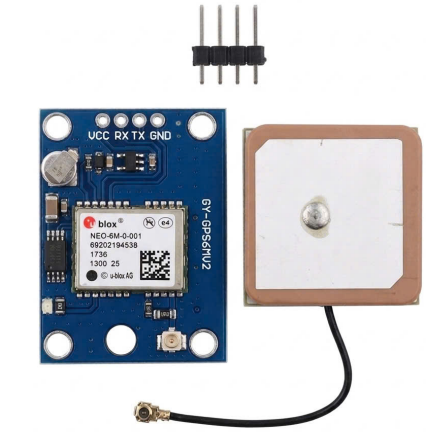


Figure 2.2. NEO-6M GPS module with external antenna

Table 2.2. NEO-6M GPS Module Parameters

Parameter	**Value**
Communication Interface	UART (TTL serial communication)
Default Baud Rate	9600 bps
Output Protocol	NMEA (default), UBX Binary
Board Supply Voltage	3.3V – 5V DC
Tracking Sensitivity	-161 dBm
Horizontal Position Accuracy	About 2.5 m
Navigation Update Rate	1 Hz (default), up to 5 Hz
Antenna Type	External antenna

2.5 A7680C LTE Cat 1 Module and Wireless Warning Communication

In the practical implementation of this project, the original SIM800L module was replaced by the SIMCom A7680C module to maintain the mobile communication function of the system under current network conditions. Although the

communication standards of the two modules are different, they play a similar role in the proposed model, namely enabling the device to exchange warning information through a mobile network.

The actual hardware of the A7680C module used in the practical model is shown in Figure 2.3. In this project, the module acts as the wireless communication unit that receives control commands from the microcontroller and transmits warning information to the user when an abnormal event is detected.

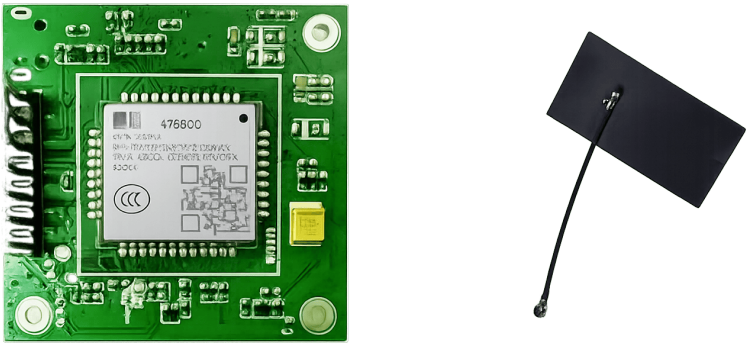


Figure 2.3. Actual hardware of the A7680C LTE Cat 1 module

According to the official SIMCom information, the A7680C is a compact Long-Term Evolution (LTE) Cat 1 communication module intended for Internet of Things (IoT) and embedded applications. It supports LTE-TDD and LTE-FDD wireless modes, provides common hardware interfaces such as UART, General-Purpose Input/Output (GPIO), and Universal Serial Bus (USB), and can be controlled by AT commands SIMCom Wireless Solutions Limited (2021, 2026). These features make the module suitable for integration into a motorcycle monitoring and anti-theft device.

In the proposed system, the A7680C module is connected to the PIC16F877A microcontroller through UART communication. The microcontroller acts as the central control unit and sends commands to the module when warning transmission is required. Whenever unusual vibration is detected or the motorcycle is determined to have moved outside the predefined safe zone, the controller activates the communication block so that warning information can be delivered

to the owner.

Compared with a local alarm only, the use of a mobile communication module makes the system more practical because the owner can still receive a warning even when staying far from the vehicle. In this way, the communication block extends the anti-theft function from on-site signaling to remote notification.

Another practical advantage of the A7680C is that it is designed for compact embedded applications and supports a package style compatible with some previous SIMCom communication modules, which makes hardware replacement more convenient in small system designs SIMCom Wireless Solutions Limited (2021). This is beneficial for the present project because the communication block can be updated while preserving the general structure of the anti-theft model.

Table 2.3. A7680C LTE Cat 1 Module Parameters

Parameter	**Value**
Supply Voltage	3.4V – 4.2V DC
Typical Voltage	3.8V DC
Maximum Data Rate	Uplink up to 5 Mbps; Downlink up to 10 Mbps
Operating Temperature	-40°C to +85°C
Dimensions	15.7 × 17.6 × 2.1 mm

The main technical specifications of the A7680C module used in this project are summarized in Table 2.3, based on the official SIMCom product information and release note SIMCom Wireless Solutions Limited (2021, 2026).

2.6 Supporting Devices in the System

In addition to the main units such as the microcontroller, GPS, and GSM modules, the system also uses several supporting devices for signal detection, information display, local warning, power regulation, and program uploading. Although these components do not perform the main processing tasks, they play an important role in making the system complete and more stable during testing and practical operation.

2.6.1. SW-420 Vibration Sensor

In this project, the SW-420 vibration sensor is used to detect unusual physical impacts on the motorcycle, such as shaking, collision, or possible tampering. When the vibration level exceeds the preset threshold, the sensor sends a signal to the microcontroller to start the warning process. This helps the system detect suspicious situations at an early stage and respond more quickly.

The SW-420 module usually provides a digital output and allows sensitivity adjustment by a potentiometer. Because of its simple structure and easy connection, it is suitable for embedded anti-theft applications.

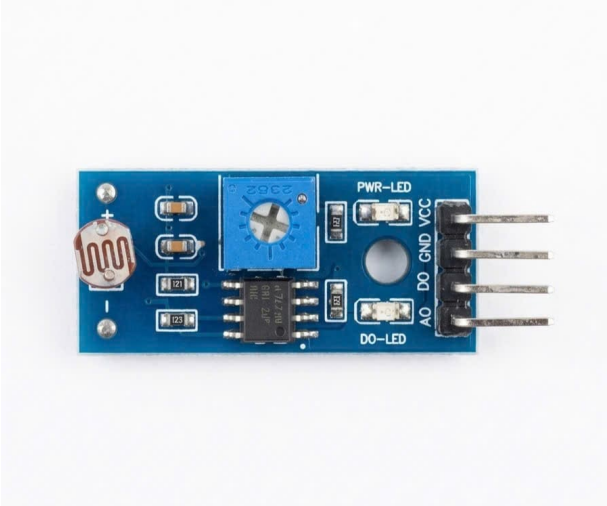


Figure 2.4. SW-420 vibration sensor used in the system

The main technical parameters of the SW-420 vibration sensor used in this project are summarized in Table 2.4.

Table 2.4. SW-420 Parameters

Parameter	**Value**
Module Name	SW-420 Vibration Sensor
Operating Voltage	3.3V – 5V DC
Output Type	Digital output
Comparator	LM393
Sensitivity Adjustment	Adjustable by potentiometer
Main Function	Detect abnormal vibration or shock

The parameters of the SW-420 vibration sensor module used in this project are summarized in Table 2.4, based on the SW-420 module documentation provided by SunFounder SunFounder (2025).

2.6.2. LCD 16x2 Display

The LCD 16x2 display serves as a direct user interface to show the basic information of the system during operation. In this project, it is primarily utilized to output real-time GPS coordinates, current system status, and diagnostic messages during the testing phase. By providing immediate visual feedback, it makes the model significantly easier to monitor, allowing the user to observe the system's behavior and troubleshoot without relying on a constant computer connection.

Hardware-wise, the LCD 16x2 is a standard alphanumeric character display featuring 2 rows and 16 characters per row. It commonly operates at a 5V logic level, ensuring seamless compatibility with most microcontrollers. It communicates with the main control board through a parallel interface, which can be configured in either 4-bit or 8-bit mode. The 4-bit mode is often preferred in such projects to save GPIO pins while still maintaining reliable and efficient screen updates.

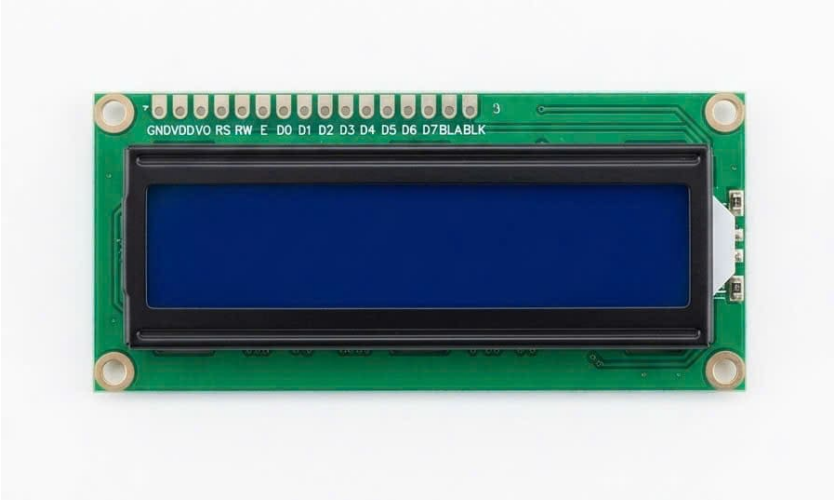


Figure 2.5. LCD 16x2 display module

The basic specifications of the LCD 16x2 display module are presented in Table 2.5, based on a standard LCD1602 module specification Shenzhen Eone Electronics (2000).

Table 2.5. LCD 16x2 Parameters

Parameter	Value
Display Format	16 × 2 characters
Operating Voltage	5V DC
Interface Mode	4-bit / 8-bit parallel

2.6.3. Buzzer Alarm

An active buzzer, as illustrated in Figure 2.6, is an audio signaling device widely utilized in embedded systems for generating acoustic alerts. Unlike a passive buzzer, which requires an external alternating signal—such as a Pulse Width Modulation (PWM) wave—to produce sound, an active buzzer features an integrated internal oscillating circuit. Consequently, its operating principle is highly straightforward, requiring only a continuous direct current (DC) voltage to function. When a simple high or low logic signal (depending on the specific module's design) is applied to its control pin, the internal oscillator activates the sound-generating element, producing a constant tone at a predetermined frequency. This characteristic significantly reduces the computational load and programming complexity for the microcontroller, making the active buzzer a highly efficient choice for basic warning mechanisms in compact hardware designs.



Figure 2.6. Active buzzer used for local warning

The buzzer specifications in Table 2.6 are adapted from a representative 5V

active buzzer datasheet with similar electrical characteristics PUI Audio, Inc. (2015).

The main electrical characteristics of the buzzer used for local warning are listed in Table 2.6.

Table 2.6. Buzzer Parameters

Parameter	Value
Operating Voltage	3.5V – 5.5V DC
Operating Current	About 30 mA @ 5V
Typical Frequency	About 2.5 kHz

2.6.4. LM2596 Power Supply Module

The LM2596 module is used as the voltage regulation unit of the system. Its main function is to step down and stabilize the input voltage so that the other modules can operate properly. In this project, the power supply block is especially important because the GSM module requires a stable supply during communication.

By using the LM2596 module, the system can reduce the risk of voltage drop and unstable operation. This improves the reliability of the whole circuit, especially when multiple modules are working at the same time.

The main specifications of the LM2596 power supply module are summarized in Table 2.7, based on the official LM2596 datasheet Texas Instruments (2026).

Table 2.7. LM2596 Parameters

Parameter	Value
Input Voltage	3.2V – 40V DC
Output Voltage	1.25V – 35V DC (adjustable)
Output Current	2A rated, up to 3A max

2.6.5. LED

LEDs are used as simple status indicators in the experimental model. In this project, they are employed to represent operating conditions such as power state, lock status, or system response during testing. Their advantages include simple interfacing, low cost, and clear visual indication for quick observation. Since the



Figure 2.7. LM2596 step-down power supply module

LEDs in this system are used only for indication, it is sufficient to consider their basic electrical characteristics, namely a typical forward voltage of about 2.0–2.2 V and a forward current of about 20 mA APEM (2017).



Figure 2.8. LEDs used for status indication

2.6.6. Push Buttons

Two push buttons are used in the experimental model as simple manual input devices. The first button is used to simulate the ACC or engine state of the motorcycle during testing. When this button indicates the engine-on condition, the system operates in normal mode, the green LED is turned on, and the current GPS coordinates can be displayed. When the button indicates the engine-off condition, the system enters anti-theft mode and stores the current coordinates as the safe-zone reference point.



Figure 2.9. Push buttons used in the experimental model

The second push button is used to silence the buzzer alarm when the user wants to stop the local warning sound. When this button is pressed, the microcontroller reads the signal as a digital input and turns off the buzzer output. Although these components are simple, they are useful for testing and user control because they allow the experimental model to simulate operating states and manage the local warning response conveniently.

2.6.7. Resistor



Figure 2.10. Resistors used in the experimental model

Resistors are passive components used in the circuit to limit current, divide voltage, and stabilize signal levels. In this project, resistors with values of 220 Ω, 1 kΩ, and 10 kΩ are used depending on the function of each circuit branch. The 220 Ω resistor is mainly used to limit the current of LEDs, while the 1 kΩ and 10 kΩ resistors are applied in signal control and logic stabilization circuits. The resistors used in the model have a typical tolerance of ±5% and a power rating of 1/4 W, which are suitable for the low-power operating conditions of the proposed system.

2.6.8. Capacitor

Capacitors are passive components used in electronic circuits to improve signal stability and support normal circuit operation. In this project, a 22 pF ceramic capacitor is used in the oscillator circuit of the PIC16F877A microcontroller. Its main purpose is to work together with the external crystal to generate a stable clock signal for the processing unit. Although it is a small component, it plays an important role in maintaining stable operation of the microcontroller and improving the reliability of the whole system.



Figure 2.11. Ceramic capacitor used in the experimental model

2.7 Communication Standards and Data Processing Principles

The operation of the proposed motorcycle tracking and anti-theft system depends not only on the hardware modules but also on the communication methods used between them. In this project, the microcontroller must receive position data from the GPS module, communicate with the GSM module, and process serial messages correctly. Therefore, UART communication, NMEA message processing, and AT-command-based control are important principles that connect the hardware part with the software algorithm.

2.7.1. UART Communication

UART is a commonly used asynchronous serial communication method in embedded systems. In this communication method, data are transmitted bit by bit without a separate clock line, so the connected devices must use the same baud rate and frame format. This makes UART suitable for communication between the PIC16F877A microcontroller and external modules such as GPS and GSM Gong et al. (2023).

In this project, UART is used as the main serial interface for exchanging data between the controller and the communication modules. In the practical implementation, the single hardware UART of the PIC16F877A is reused sequentially:

it receives coordinate strings from the GPS module and then transmits command strings to the mobile communication module for SMS warning transmission.

2.7.2. Single Hardware UART Arrangement for GPS Reception and GSM Transmission

PIC16F877A has only one hardware UART, so the final implementation of this project reuses that UART for both GPS reception and GSM warning transmission at different stages of operation. In this arrangement, the GPS module sends NMEA coordinate data to the receive pin of the microcontroller. After the data are processed, the transmit pin of the microcontroller is used to send SMS command strings to the mobile communication module. This solution was chosen because the software UART approach was less stable in practical testing, while the single hardware UART provided more reliable communication in the experimental model.

2.7.3. NMEA-0183 Message Format

The GPS module outputs positioning information in the form of serial text messages that follow the NMEA-0183 standard. This standard defines sentence-based data structures used by GPS receivers to transmit navigation information such as time, latitude, longitude, speed, and signal status Langley (1995).

In practical operation, the microcontroller receives these NMEA strings and extracts the useful fields needed for the project, especially latitude and longitude. These coordinate values are then used for LCD display, safe-zone monitoring, and warning generation when abnormal movement is detected.

2.7.4. AT Commands for SMS Transmission

To send warning messages, the mobile communication module is driven by AT command strings through UART communication. In the practical implementation of this project, after GPS data have been received and processed, the microcontroller uses its UART transmit line to forward the required SMS command sequence to the mobile communication module. Whenever abnormal vibration is detected or the motorcycle moves outside the safe zone, the controller sends the corresponding warning command sequence for message transmission.

CHAPTER 3. HARDWARE DESIGN

3.1 Overall System Block Diagram

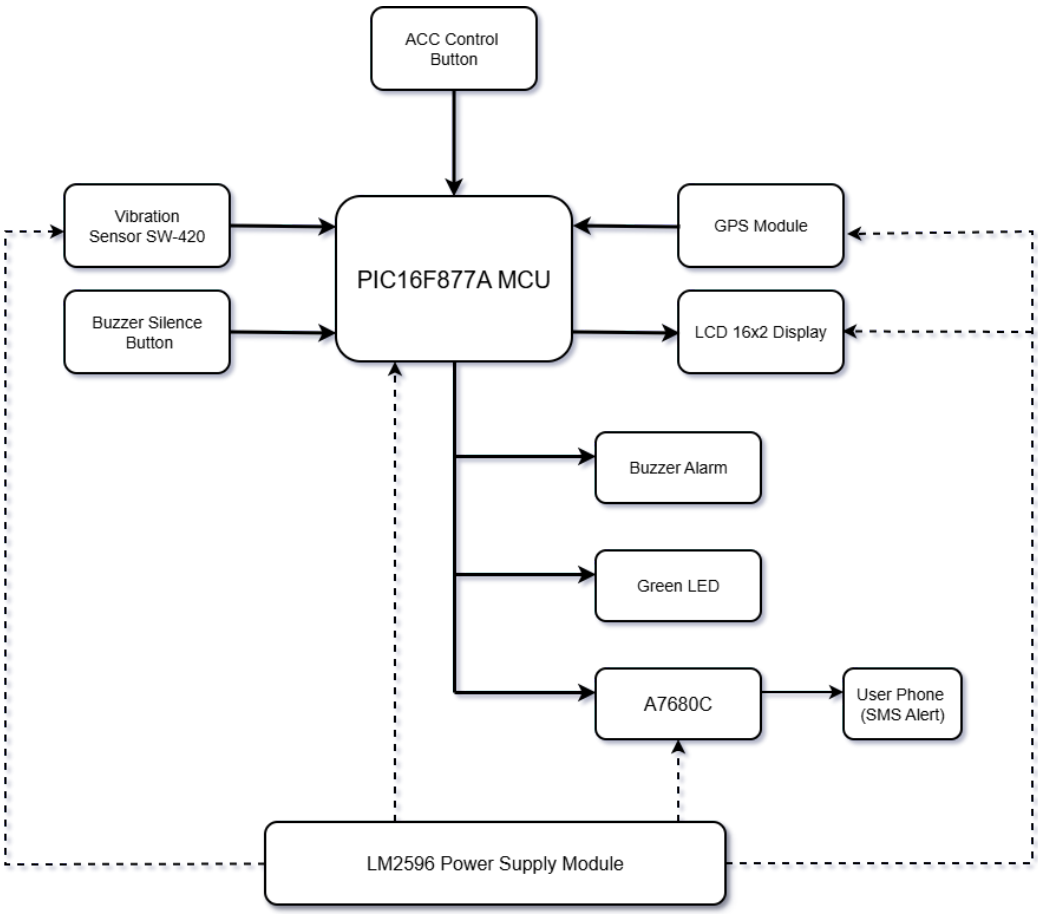


Figure 3.1. Block diagram of the proposed motorcycle tracking and anti-theft system

The overall block diagram of the proposed motorcycle tracking and anti-theft system is shown in Figure 3.1. In this design, the PIC16F877A microcontroller serves as the central control unit and coordinates the operation of all modules in the system. An ACC control button is used in the experimental model to simulate the engine ON/OFF state and determine the working mode of the system, including the normal mode and the parking or anti-theft mode. At the same time, the SW-420 vibration sensor is used to detect abnormal physical impacts, while a second push button is used to silence the buzzer alarm when required by the user.

The GPS module supplies location data to the microcontroller through UART

communication, while the LCD 16x2 display is used to show the current system status and coordinate information. When an abnormal condition is detected, the microcontroller can activate the buzzer alarm and control the green LED for status indication. In addition, the mobile communication module is used to send warning messages to the user's phone in the form of SMS alerts. To ensure stable operation of the whole system, the LM2596 power supply block provides regulated power for the main modules.

In general, the block diagram shows how the input, processing, output, communication, and power blocks are connected together to form a complete motorcycle monitoring and anti-theft system. This structure provides the foundation for the detailed hardware design and circuit implementation presented in the following sections.

3.2 Power Supply Block

3.2.1. Schematic

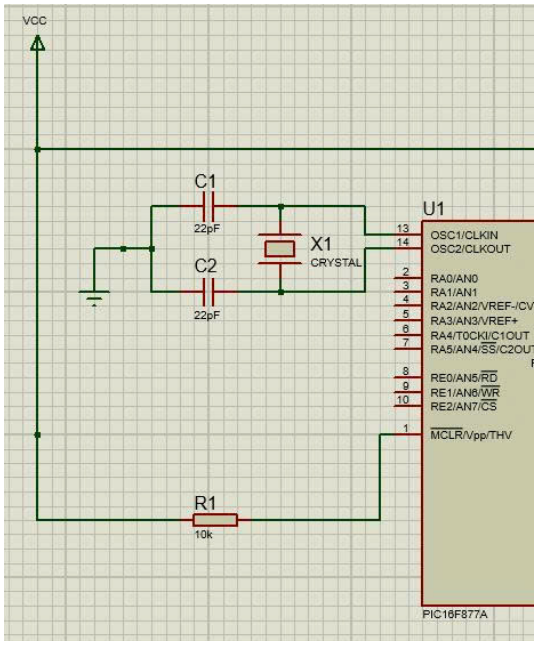


Figure 3.2. Power supply and Reset circuit for PIC16F877A in simulation

Figure 3.2 illustrates the power supply section of the hardware design. In the Proteus simulation, the circuit is powered by an ideal regulated DC source (VCC line). However, for the practical implementation, the LM2596 buck converter

module is utilized to step down the input voltage to the required stable operating voltages for the entire system.

3.2.2. Operating Principle and Implementation

The power supply block is responsible for providing stable operating voltages for the main units of the proposed system. In this project, the LM2596 module is used as the main voltage regulator because it offers adjustable output voltage, sufficient output current, and stable operation for embedded applications. The regulated 5V supply is used for the PIC16F877A microcontroller, the LCD 16x2 display, and the sensing circuits.

In addition to the 5V line, the mobile communication module requires a lower and more stable operating voltage. According to the specifications presented in Chapter 2, the A7680C module operates in the range of 3.4V to 4.2V, with a typical value of 3.8V. Therefore, the power supply design must ensure that the communication block receives a suitable and stable voltage during operation, especially when warning transmission is activated.

For the experimental model, the system is powered by an external battery source. However, in practical motorcycle installation, the device can be supplied from the motorcycle battery through the LM2596 regulator. In this case, the regulator can step down and stabilize the input voltage to the required levels for the control, display, sensing, and communication blocks. This arrangement improves the practicality of the proposed anti-theft system and helps maintain longer operating time in real use.

3.3 Control Block

3.3.1. Schematic and Hardware Design

Figure 3.3 presents the control block, which serves as the core of the system centered around the PIC16F877A microcontroller. To ensure reliable execution of the control program, the Microcontroller Unit (MCU) is configured with a basic hardware setup. A 20MHz external crystal oscillator is connected with two 22 pF capacitors to provide a stable clock signal, which is strictly required for accurate UART baud rate generation. Additionally, a 10 kΩ pull-up resistor is connected to the Master Clear Reset (MCLR) pin to prevent unintentional resets and maintain a stable operating state.

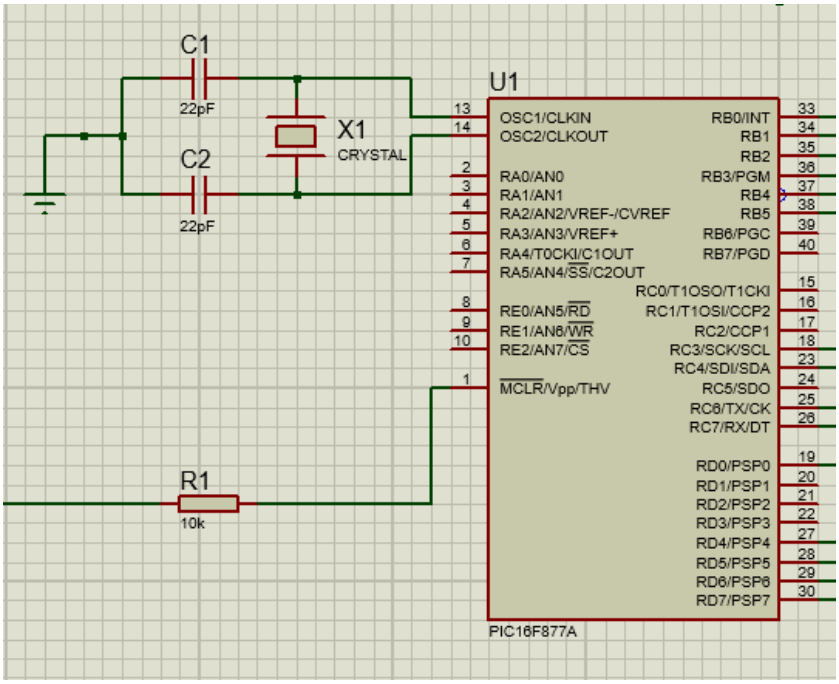


Figure 3.3. Schematic of the control block in Proteus

3.3.2. Operating Principle

The ACC control button is used to simulate the engine ON/OFF state in the experimental model. When the ACC input indicates the engine-on condition, the system operates in normal mode. In this state, the green LED is turned on and the current GPS coordinates can be displayed. When the ACC input indicates the engine-off condition, the controller stores the current position as the safe-zone reference and activates the anti-theft monitoring process.

If an abnormal event is detected, the control logic activates the buzzer alarm and sends the required SMS warning command sequence to the mobile communication module.

3.4 Input and Sensing Block

3.4.1. Schematic

Figure 3.4 shows the input and sensing block of the proposed system. This block includes the ACC control button, the SW-420 vibration sensor input, and the buzzer silence button.

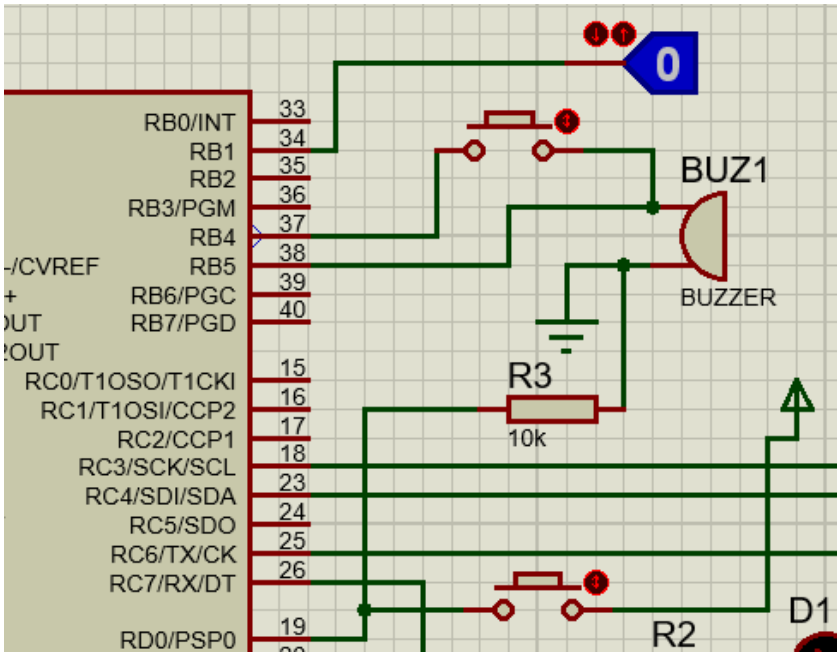


Figure 3.4. Schematic of the input and sensing block in Proteus

3.4.2. Operating Principle

The ACC control button is used to simulate the engine ON/OFF state in the experimental model. When the button indicates the engine-on condition, the system operates in normal mode. In this state, the green LED is turned on and the current GPS coordinates can be displayed. When the button indicates the engine-off condition, the controller stores the current position as the safe-zone reference and activates the anti-theft monitoring process.

The SW-420 input is used to detect abnormal physical impact on the motorcycle, such as shaking or tampering. If such an event is detected, the microcontroller immediately starts the warning procedure.

The second push button is used to turn off the buzzer alarm when the user needs to silence the local warning sound. The actual detection of movement outside the safe zone is based on GPS position comparison, not on this push button input.

3.4.3. Design

The input and sensing block is designed with simple digital input signals so that the microcontroller can read them directly. This block combines the ACC control button for mode simulation, the vibration sensor for intrusion detection, and the

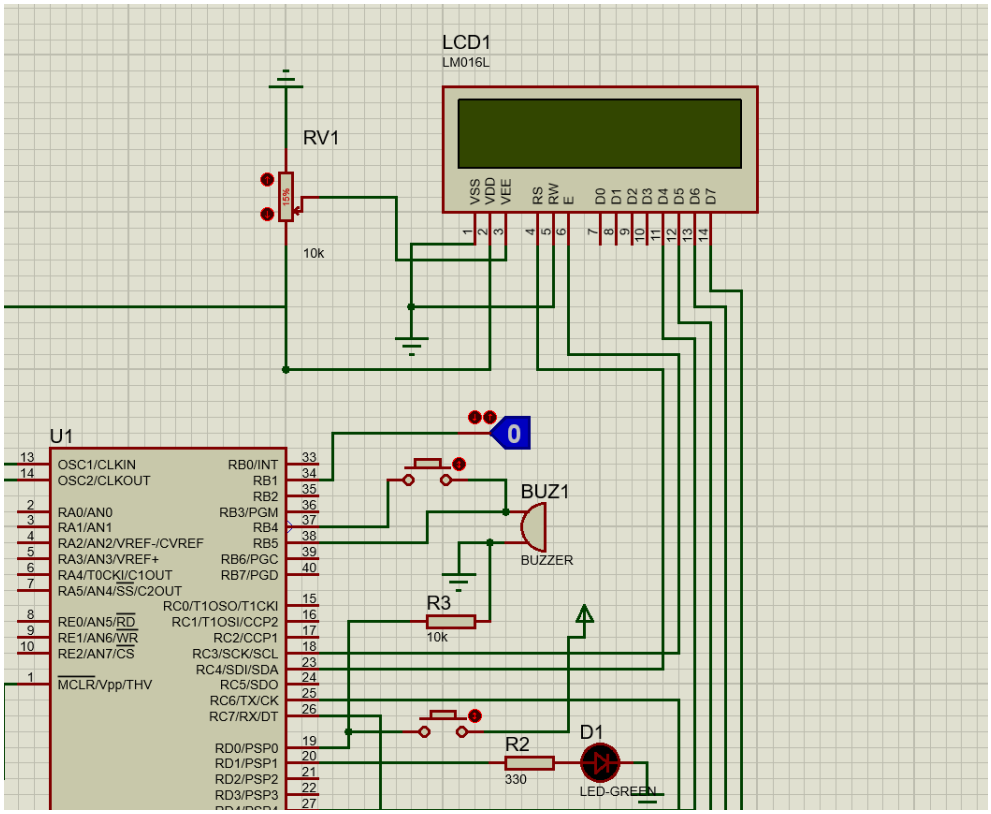


Figure 3.5. Schematic of the display and alarm block in Proteus

buzzer silence button for user control. Such an arrangement makes the system easier to implement in both simulation and practical testing, while helping the software distinguish between operating-state input, abnormal physical intrusion, and manual buzzer control.

3.5 Display and Alarm Block

3.5.1. Schematic

Figure 3.5 presents the display and alarm block of the system. This part includes the LCD 16x2 display, the contrast adjustment potentiometer, the buzzer alarm, and the green LED indicator.

3.5.2. Operating Principle

The LCD 16x2 is used to display system information such as operating status and GPS coordinates. This helps the user monitor the system directly during testing and operation. The buzzer is used as the local warning device and is activated whenever an abnormal event is detected. The green LED is used as a

status indicator to represent the engine-on condition in the experimental model. When the ACC control button indicates the engine-on state, the LED is turned on. When the system changes to the engine-off or anti-theft state, the LED is turned off.

3.5.3. LCD, Buzzer, and LED in the Proposed System

The LCD is connected to the microcontroller in 4-bit mode in order to save I/O pins while still providing sufficient display capability. The buzzer is driven by the microcontroller to generate an immediate sound warning, and the LED is connected through a resistor to limit current and protect the component.

3.5.4. Design

In the Proteus design, the LCD is supported by a potentiometer for contrast adjustment. The LED is connected in series with a 330Ω resistor for current limiting, while the buzzer is connected as an alarm output controlled by the microcontroller. This arrangement is simple, practical, and suitable for a small embedded anti-theft model.

3.6 GPS and GSM Interface Block

3.6.1. Schematic

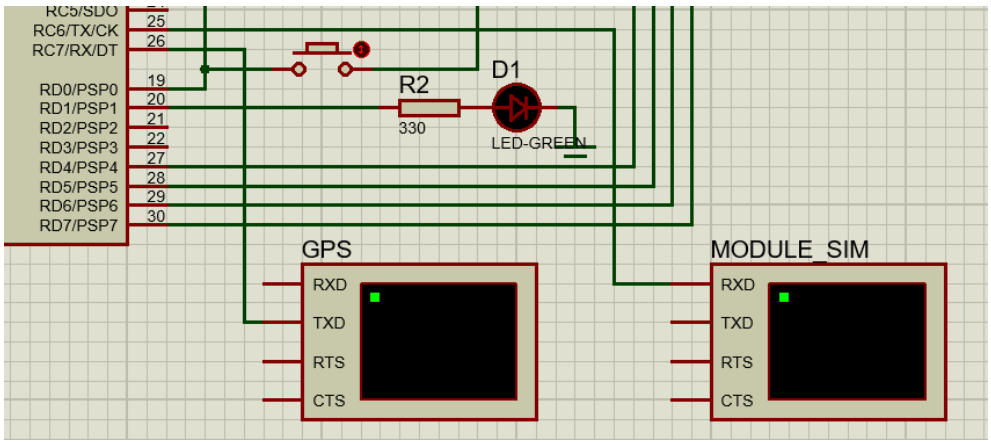


Figure 3.6. Schematic of the GPS and GSM interface block in Proteus

Figure 3.6 shows the GPS and GSM interface section of the system. In the simulation model, the serial communication lines are arranged so that the transmit

pin of the GPS terminal is connected to the receive pin of the PIC16F877A, allowing the microcontroller to receive coordinate data. In addition, the transmit pin of the PIC16F877A is connected to the receive pin of the mobile communication terminal so that SMS command strings can be sent during warning operation.

3.6.2. Operating Principle

The GPS interface is used to provide coordinate data for position monitoring. In this design, the GPS module sends NMEA data through its transmit line to the receive line of the microcontroller. These data are then used for coordinate display, safe-zone comparison, and warning decision making. In this project, the detection of unauthorized movement is based on GPS position data rather than on a manual trigger input.

The mobile communication interface is used to transmit warning messages through SMS. After the required GPS data have been received and processed, the microcontroller uses its UART transmit line to send the required SMS command sequence to the mobile communication module so that warning information can be delivered to the user's phone.

3.6.3. Design

The communication block is designed by reusing the single hardware UART of the PIC16F877A for two serial tasks at different stages of operation. In this arrangement, the GPS module provides NMEA coordinate data to the receive line of the microcontroller, and after the data are processed, the transmit line of the microcontroller is used to send SMS command strings to the mobile communication module.

This design was selected because the software UART solution was found to be less stable in practical testing. By using only the hardware UART, the experimental model achieves more reliable serial communication while still supporting both GPS monitoring and SMS warning transmission.

CHAPTER 4. SOFTWARE DESIGN

4.1 Overall Program Flowchart

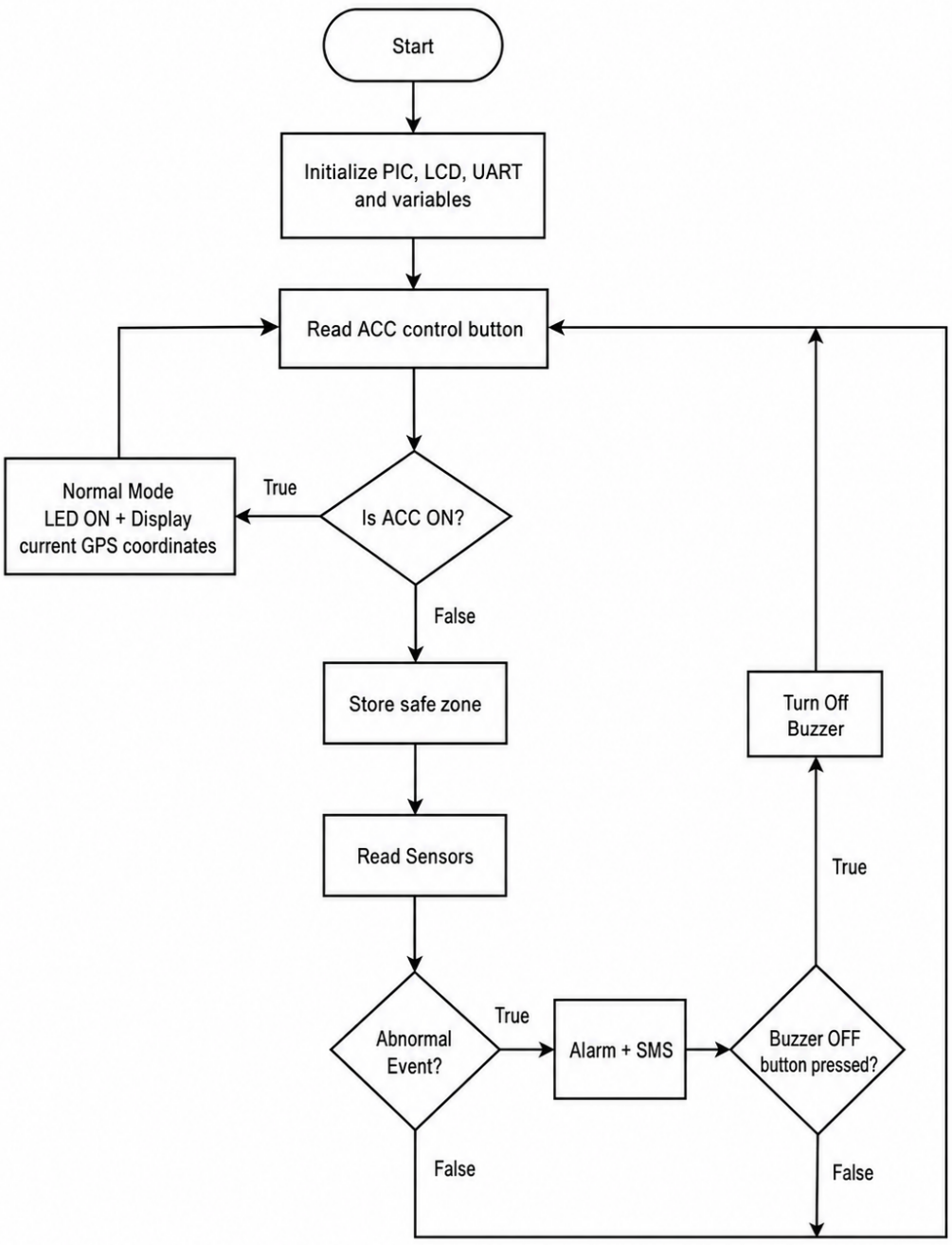


Figure 4.1. Overall program flowchart of the proposed anti-theft system

Figure 4.1 illustrates the overall program flow of the proposed anti-theft system. After start-up, the microcontroller initializes the PIC control unit, LCD display, UART communication, and the variables required for operation. The system then reads the ACC control button state to determine the current operating mode.

If the ACC input indicates the engine-on condition, the system operates in normal mode. In this state, the anti-theft function is disabled, the green LED is turned on, the buzzer is turned off, and the current GPS coordinates can be displayed. If the ACC input indicates the engine-off condition, the system enters anti-theft mode. At this stage, the current GPS position is stored as the safe-zone reference point only when SafeZoneSaved has not been set yet.

After that, the microcontroller reads the sensor signals and checks for abnormal events caused by unusual vibration or the motorcycle moving outside the predefined safe zone. When an abnormal event is detected, the buzzer alarm is activated and an SMS warning is sent to the user.

Finally, the program checks the buzzer silence button state. If this button is pressed, the buzzer is turned off. Otherwise, the system continues monitoring and returns to the main loop.

4.2 GPS Data Processing and Coordinate Extraction

Figure 4.2 presents the GPS data processing flow of the proposed system. The procedure begins when the microcontroller receives an NMEA string from the GPS module through its hardware UART receive line.

If the sentence is valid, the program extracts the coordinate information, especially latitude and longitude, from the received data. These values are then converted into the required format so that they can be used more easily in the monitoring process. After the conversion step, the system updates the current position of the motorcycle and makes this information available for display and further distance calculation.

If the received sentence is not valid, the system does not update the position with incorrect data. Instead, it keeps using the last known valid position stored in memory. This approach helps maintain basic monitoring capability when the GPS signal is temporarily lost or unstable. In this way, the system can process GPS data more reliably and support the anti-theft function more effectively.

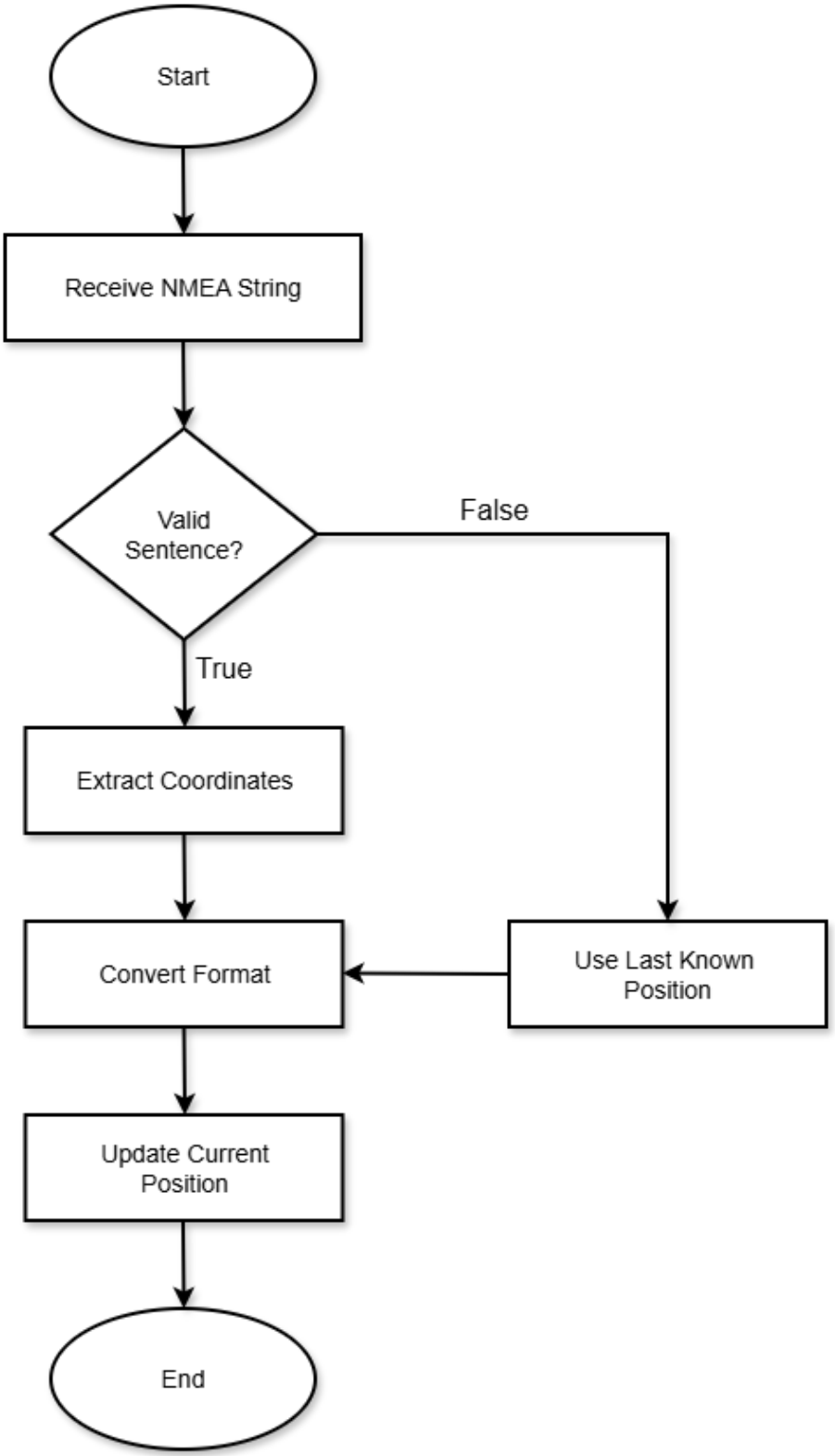


Figure 4.2. GPS Data Processing Flowchart

4.3 Anti-Theft and SMS Warning Flowchart

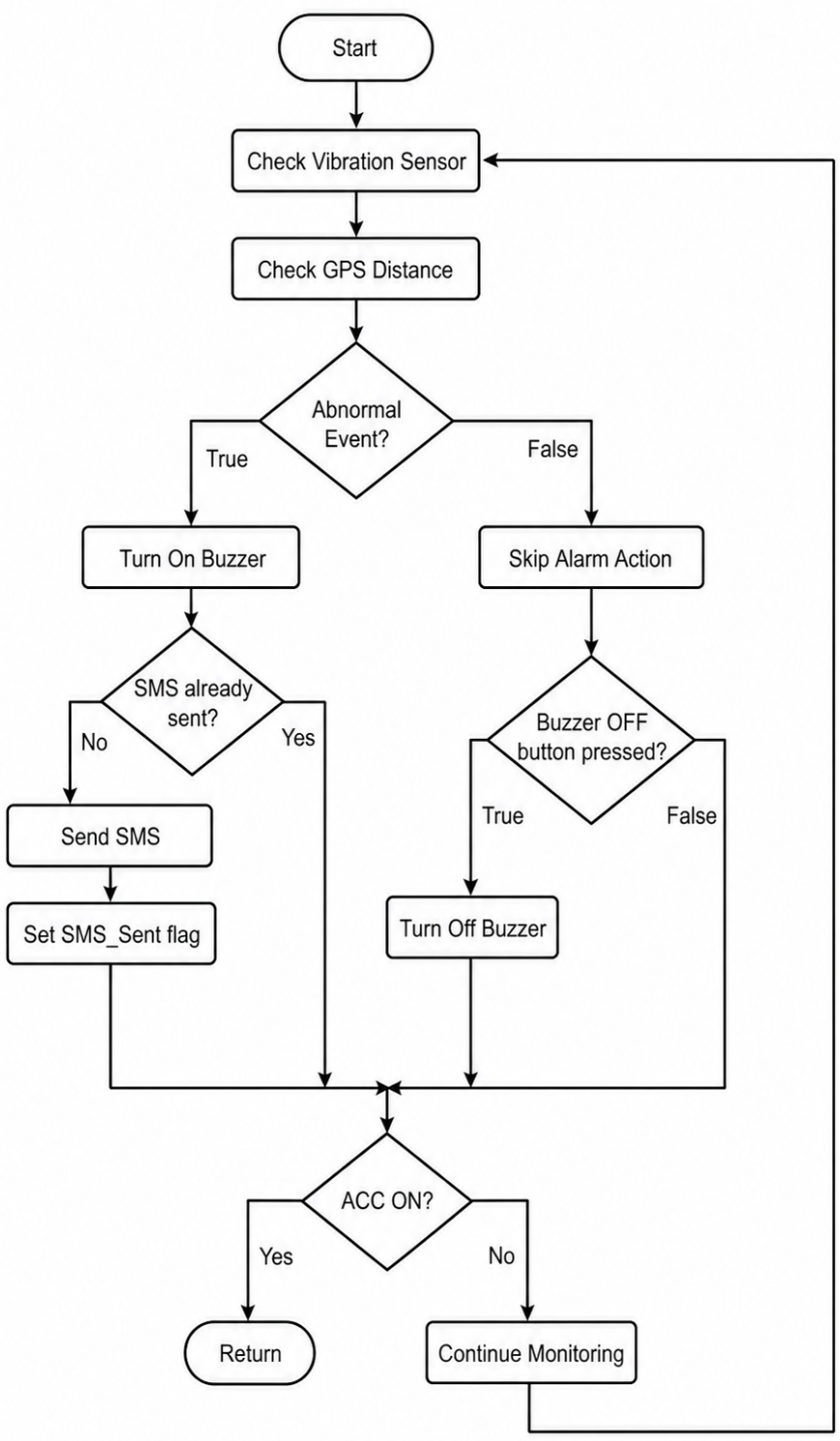


Figure 4.3. Anti-Theft and SMS Warning Flowchart

Figure 4.3 describes the anti-theft warning process of the proposed system. In this procedure, the microcontroller first checks the vibration sensor and then evaluates the GPS distance from the stored safe-zone center. Based on these two conditions, the program determines whether an abnormal event is present. An abnormal event may be caused by unusual vibration or by the motorcycle moving outside the predefined safe zone.

When an abnormal event is detected, the system turns on the buzzer to provide an immediate local alarm. At the same time, the program checks whether a warning message has already been sent. If not, the system creates the required warning content and sends the SMS command sequence to the mobile communication module through the UART transmit line, then sets the SMS Sent flag to prevent repeated transmission for the same event.

If no abnormal event is detected, the program skips the alarm action and continues to the next control steps. After that, the system checks the buzzer silence button state. When this button is pressed, the buzzer is turned off according to the user request. Finally, the program checks the ACC state. If ACC is turned on again, the system exits anti-theft mode and returns to the normal operating condition. In this way, the warning process can operate continuously while still avoiding repeated SMS alerts and allowing the user to silence the buzzer when needed.

CHAPTER 5. IMPLEMENTATION AND TESTING

5.1 PCB

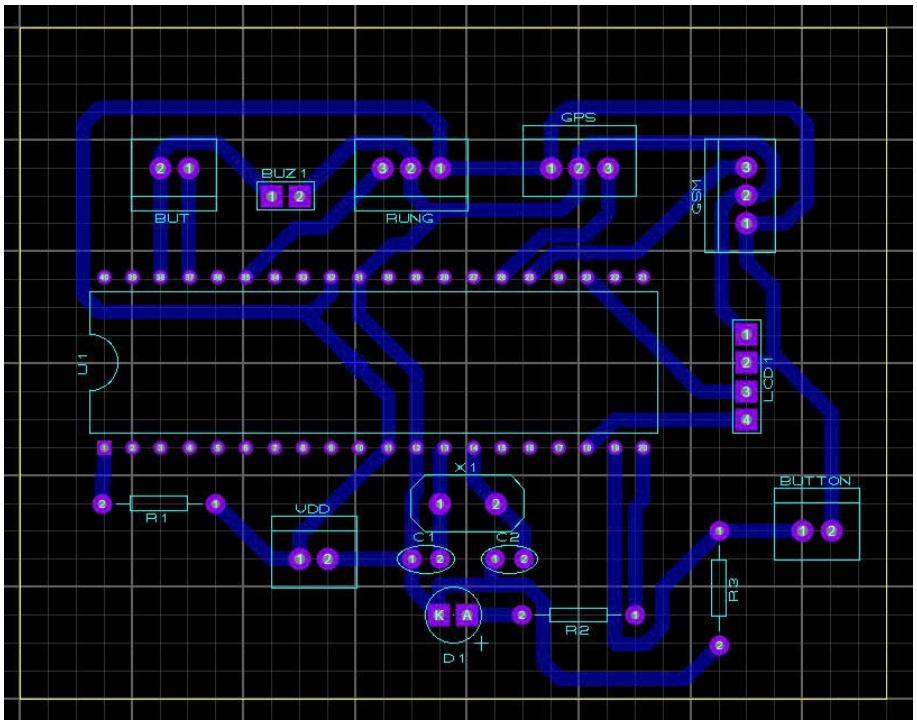


Figure 5.1. PCB layout result in Proteus

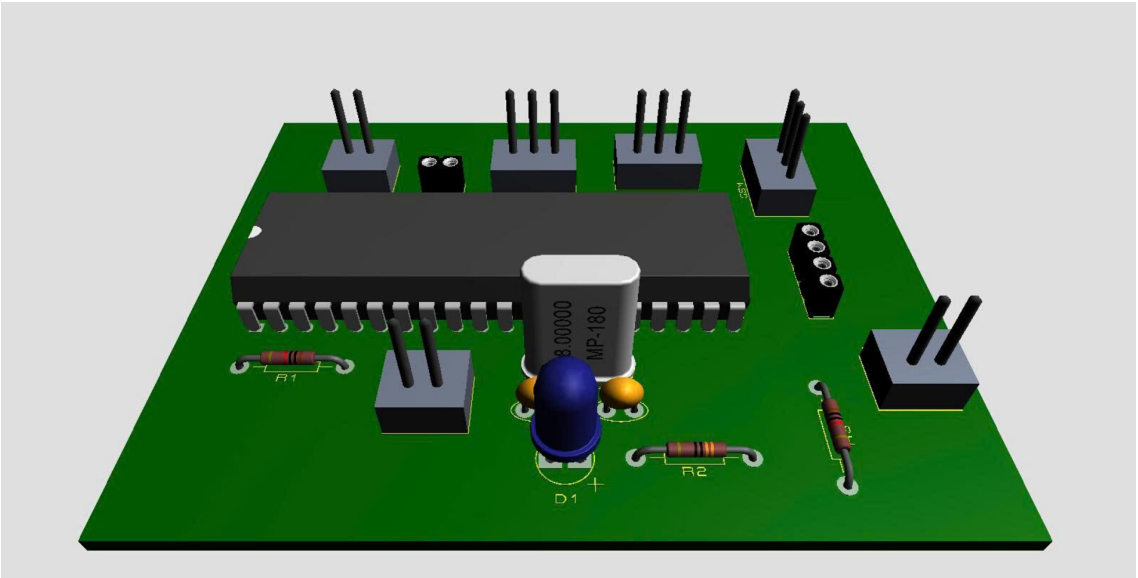


Figure 5.2. 3D visualizer result of the designed PCB

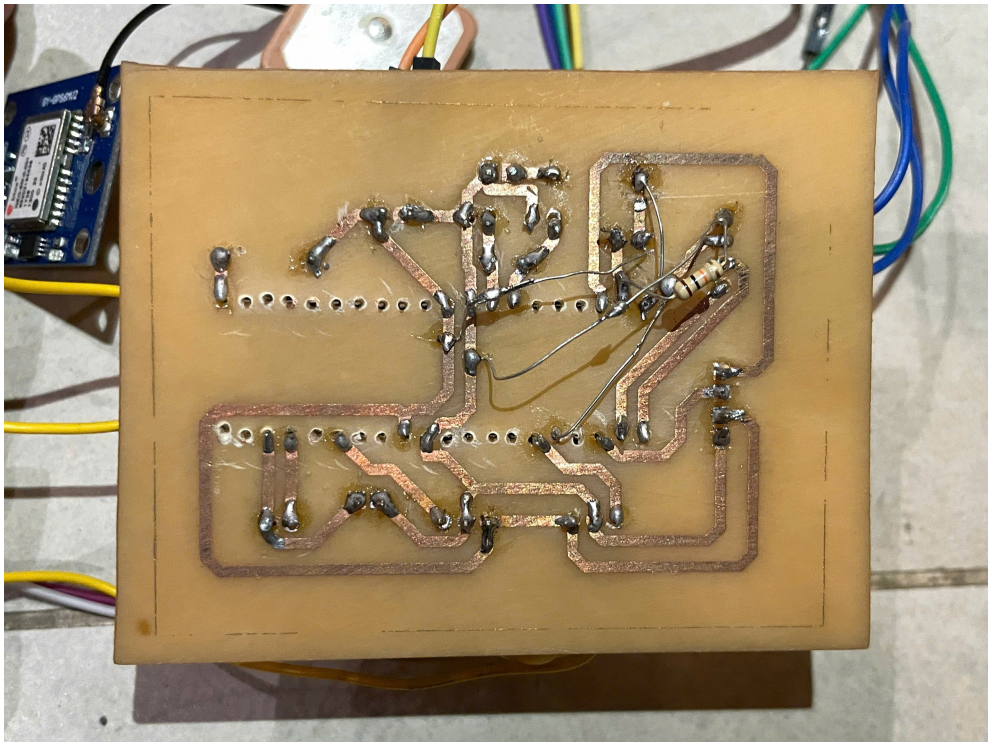


Figure 5.3. PCB after construction

Figure 5.1 shows the PCB layout result of the proposed system in Proteus. After completing the schematic design, the components were arranged and the signal traces were routed to form a compact PCB suitable for the practical model. The layout was organized to simplify the connections between the PIC16F877A microcontroller and the main peripheral blocks such as GPS, buzzer, vibration sensor, and communication module.

Figure 5.2 presents the 3D visualizer result of the designed PCB. This view provides a clearer representation of the physical arrangement of the components on the board, including the microcontroller, crystal oscillator, capacitors, LED, resistors, and connection headers. It also helps evaluate the overall placement before the fabrication stage.

Figure 5.3 shows the PCB after construction and soldering. At this stage, the copper board was etched, drilled, and assembled with the required through-hole components for the main control circuit. This result represents the transition from PCB design to real hardware implementation and confirms that the designed traces can be physically fabricated for practical use.

5.2 Complete Prototype

Figure 5.4 presents the complete prototype of the proposed motorcycle positioning and anti-theft system. After the PCB was fabricated and soldered, the main external modules were connected to form the full experimental model, including the PIC16F877A control board, LCD 16x2 display, NEO-6M GPS module, A7680C communication module, LM2596 power supply module, and battery source.

In the assembled model, the PIC16F877A microcontroller serves as the central processing unit and is mounted on the main PCB together with the crystal oscillator, LED indicator, two push buttons, and supporting components.

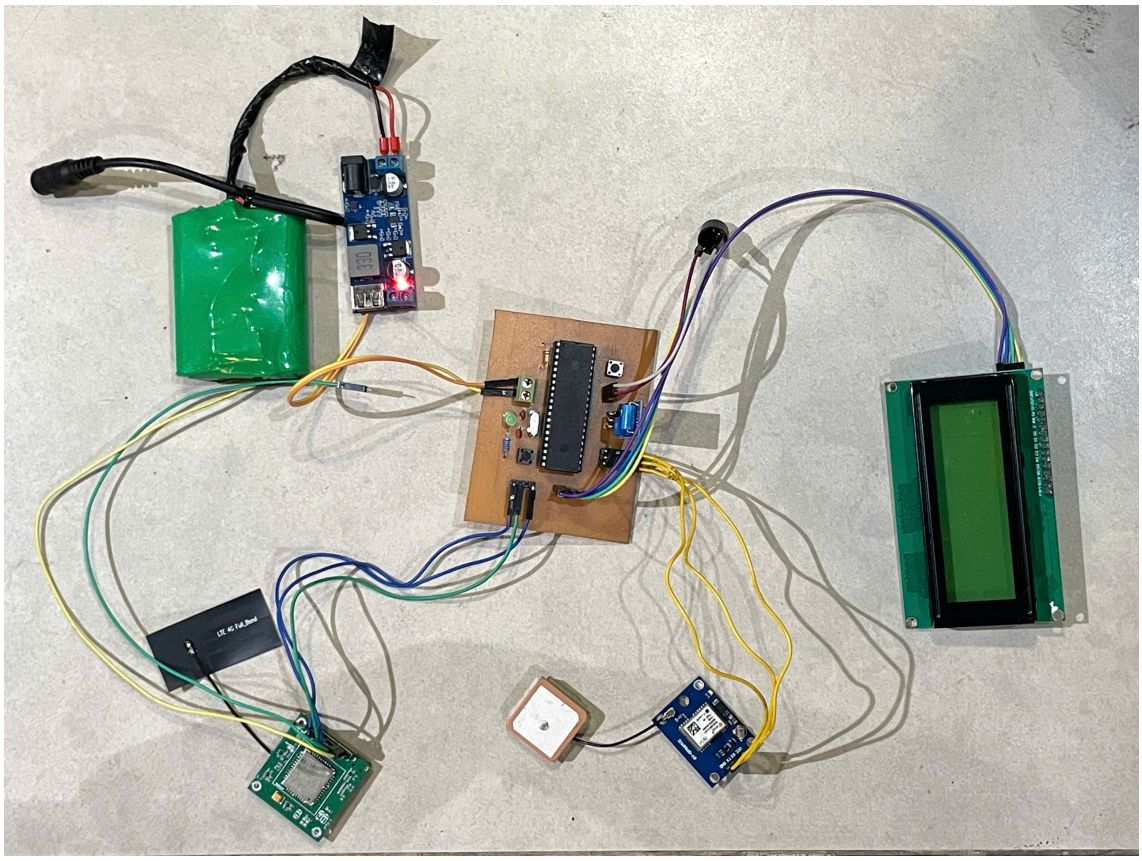


Figure 5.4. Complete prototype of the proposed anti-theft system

Overall, the complete prototype demonstrates that the designed hardware can be assembled into a working embedded model with all major functional blocks connected together. This stage provides the physical platform for subsequent testing, debugging, and performance evaluation of the proposed anti-theft system.

5.3 Practical Testing

5.3.1. Test GPS Coordinate Display in Normal Mode

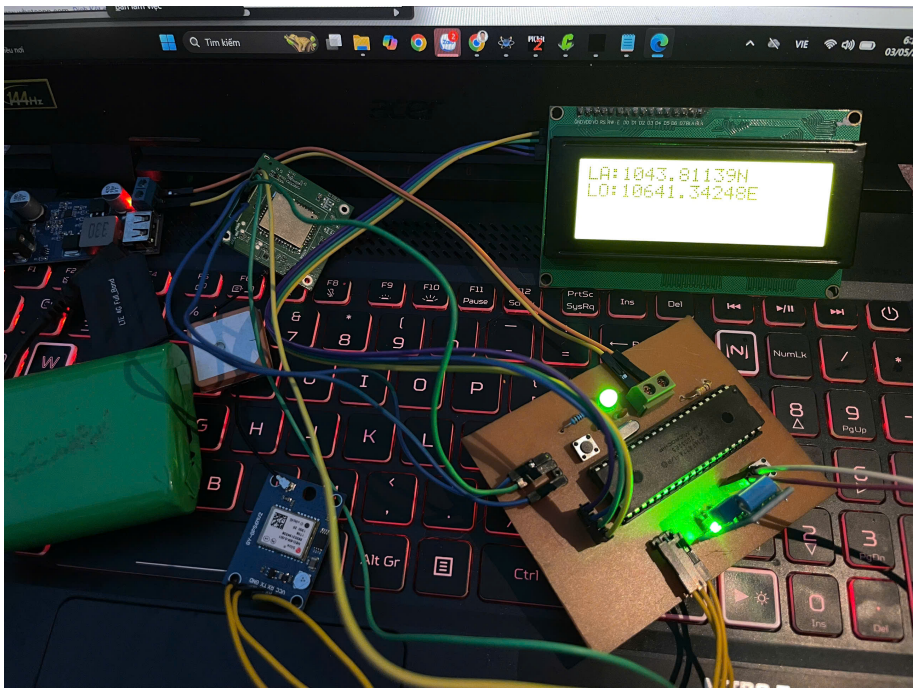


Figure 5.5. Practical testing result in normal mode with GPS coordinate display

Figure 5.5 shows the first practical testing condition of the proposed system. In this test, the hardware prototype was powered on and the ACC control button was set to represent the engine-on state. In this condition, the green LED indicator was turned on, showing that the system was operating in normal mode.

At the same time, the GPS module received the current position and transmitted the coordinate data to the PIC16F877A microcontroller. The LCD screen then displayed the latitude and longitude values of the current location. This result confirms that the proposed system can successfully receive GPS data and display the current coordinates on the LCD during the normal operating condition.

This is the initial operating stage of the system and serves as the basis for the following anti-theft process. The obtained result indicates that the GPS acquisition and display functions were implemented correctly in the practical hardware model.

5.3.2. Test Activation of Anti-Theft Mode

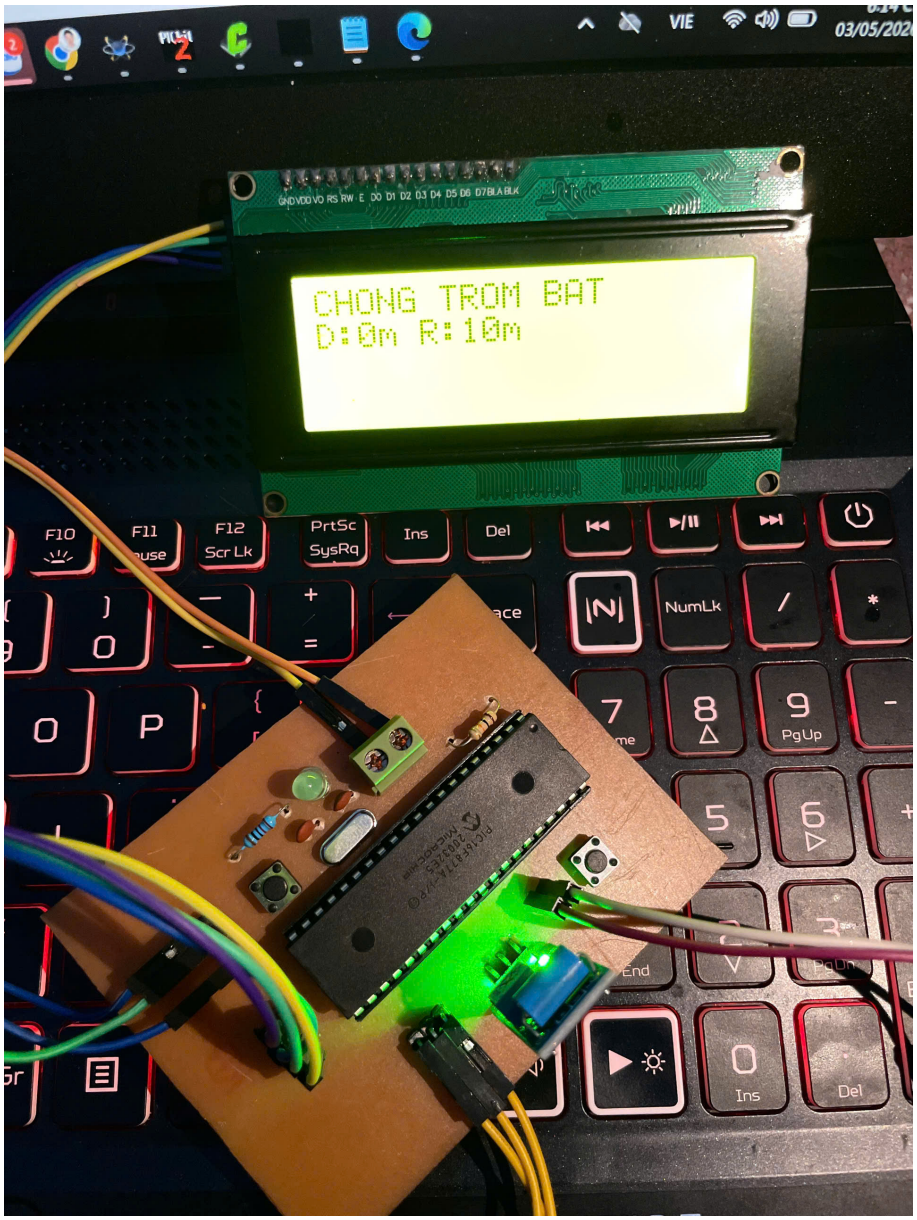


Figure 5.6. Practical testing result when anti-theft mode is activated

Figure 5.6 presents the second practical testing condition, in which the motorcycle is assumed to be parked. In this test, the ACC control button was changed to the engine-off state to simulate that the owner had turned off the motorcycle engine after parking. At this moment, the anti-theft mode was activated and the system entered the monitoring state.

The LCD displayed the anti-theft message together with the measured distance and the predefined safe radius of 10 m. In the shown result, the distance

value was 0 m, which means that the current position still matched the stored safe-zone reference point. This confirms that the system had already stored the current GPS position as the safe-zone center and was ready to detect any abnormal movement.

This testing result shows that the proposed system can correctly switch from normal operation to anti-theft mode and prepare the monitoring function after the vehicle is parked.

5.3.3. Test SMS Warning When the Vehicle Moves Outside the Safe Zone



Figure 5.7. Practical testing result of SMS warning transmission when the safe zone is violated

Figure 5.7 illustrates the third practical testing condition. In this test, the monitored position was changed so that the calculated distance exceeded the prede-

fined safe-zone radius of 10 m. This situation simulated the case in which the motorcycle was moved away from its parked position without authorization.

When the safe-zone limit was exceeded, the system immediately entered the warning process. The LCD displayed the message indicating that the SMS warning was being sent. At the same time, the mobile communication module was triggered by the microcontroller to transmit the warning message to the owner's phone.

This result verifies that the proposed system can detect unauthorized movement outside the safe zone and generate a remote warning to the user. Therefore, the practical test confirms that the GPS-based anti-theft warning function operated successfully in the experimental prototype.

5.3.4. Discussion of Testing Results

From the three practical tests, it can be observed that the main functions of the proposed system operated according to the intended design. In normal mode, the system successfully displayed the current GPS coordinates. When the motorcycle was parked, the anti-theft mode was activated and the safe-zone monitoring process was prepared correctly. When the monitored position moved outside the 10 m safe zone, the system generated an SMS warning as expected. These results demonstrate that the proposed hardware prototype can perform the essential functions of motorcycle tracking and anti-theft warning in practical operation.

CHAPTER 6. CONCLUSION

6.1 Advantages

This project has shown that a compact embedded anti-theft system can be developed by combining the PIC16F877A microcontroller with GPS positioning, vibration detection, and mobile warning communication. One important advantage of the proposed model is that it can support both local and remote warning functions. The buzzer provides an immediate on-site alarm, while the SMS function helps notify the owner when the motorcycle is moved abnormally or subjected to suspicious vibration.

Another advantage is that the system is relatively simple in structure and practical for a student-level embedded project. The selected components are common, easy to interface, and suitable for implementing the main functions of monitoring, warning, and position handling. In addition, the use of a geofencing method and last known position storage helps improve the reliability of the system under basic real operating conditions.

From the implementation point of view, the project also provides useful experience in circuit design, PCB layout, serial communication, data processing, and embedded programming. Therefore, the project is not only meaningful in practical application but also valuable in terms of learning and technical development.

6.2 Disadvantages

Besides the achieved results, the project still has several limitations. First, the accuracy of movement detection depends on the GPS signal quality. In environments such as indoor parking areas, basements, or locations with many obstacles, the GPS signal may become weak or unstable, which can affect coordinate accuracy and warning reliability.

Second, the current system is mainly designed for basic anti-theft monitoring and warning. It does not include a mobile application, a web-based tracking platform, or a more advanced user interface. As a result, the interaction between the user and the system is still limited to LCD display and SMS warning messages.

Another limitation is that the hardware model is still relatively simple and

mainly intended for academic implementation. The PCB layout and practical assembly may still need further optimization in terms of compactness, wiring quality, power stability, and overall durability for long-term real-world use.

6.3 Development

In the future, the project can be improved in several directions. One possible development is to optimize the hardware design so that the system becomes smaller, more stable, and more suitable for practical installation on a motorcycle. The communication and power blocks can also be refined to improve reliability during long-term operation.

Another development direction is to expand the user interface and monitoring capability. Instead of relying only on SMS messages, the system can be integrated with a mobile application or a web platform to support easier location tracking, warning history, and user interaction. This would make the anti-theft system more convenient and modern.

In addition, the warning logic can be enhanced by combining more sensors or more intelligent decision rules. For example, the system may verify abnormal conditions over multiple samples, combine vibration and position information more effectively, or apply additional filtering methods to reduce false alarms. These improvements would help the proposed system become more accurate, practical, and suitable for real deployment.

Another practical development is to integrate the power supply with the motorcycle battery so that the anti-theft device can maintain continuous operation for a longer time in real use.

REFERENCES

APEM (2017). Q10 series panel mount led indicators. Data Sheet Q10_VOY1704US, APEM. Accessed: 2026-04-14.

Gong, J., Guo, W., and Sun, W. (2023). Uart communication protocol frame format explanation and application. *Applied and Computational Engineering*, 14:47–56.

Langley, R. B. (1995). Nmea 0183: A gps receiver. *GPS World*, 6(7):54–57.

Microchip Technology Inc. (2003). *PIC16F87XA Data Sheet: 28/40/44-Pin Enhanced Flash Microcontrollers*. Microchip Technology Inc. Accessed: 2026-04-14.

Naik, R. K. and Nair, B. B. (2019). Improving gps based distance measurement accuracy using machine learning: an empirical study. In *2019 International Conference on Communication and Electronics Systems (ICCES)*, pages 1294–1299.

PUI Audio, Inc. (2015). *AI-1223-TWT-5V-2-R Data Sheet*. PUI Audio, Inc. Accessed: 2026-04-14.

Rao, A., Vaidya, A., and Narayanan, Y. (2022). Faculty attendance system using geofence, face-recognition and admin portal. *Turkish Journal of Computer and Mathematics Education*, 13(3):819–829.

Shenzhen Eone Electronics (2000). *1602A-1 LCD Module Specification*. Shenzhen Eone Electronics. Version 1.0. Accessed: 2026-04-14.

SIMCom Wireless Solutions Limited (2021). SIMCom Launched a New Ultra-Small CAT.1 Module A7680C. `https://www.simcom.com/news_view-159.html`. Official news release. Accessed: 2026-04-16.

SIMCom Wireless Solutions Limited (2026). *A7680C Product Information*. SIMCom Wireless Solutions Limited. Official product page. Accessed: 2026-04-16.

SunFounder (2025). Vibration Sensor Module (SW-420). `https://docs.sunfounder.com/projects/umsk/en/latest/01_components_basic/24-component_vibration.html`. SunFounder Universal Maker Sensor Kit documentation. Accessed: 2026-04-14.

Texas Instruments (2026). *LM2596 Product Page*. Texas Instruments. Official product page. Accessed: 2026-04-17.

u-blox AG (2011). *NEO-6: u-blox 6 GPS Modules*. u-blox AG. Data Sheet. Accessed: 2026-04-14.

u-blox AG (2023). u-blox 8 / u-blox m8 receiver description: Including protocol specification v15–20.30, 22–23.01. Manual UBX-13003221, u-blox AG.

APPENDIX A. CODDE

```
#include <16F877A.h>
#fuses HS,NOWDT,NOPROTECT,NOLVP,PUT
#use delay(clock=20000000)

#include <math.h>

// LCD I2C: RC4 = SDA, RC3 = SCL
#use i2c(master, sda=PIN_C4, scl=PIN_C3, FAST)

// UART dung chung:
// RC7/RX nhan GPS TX
// RC6/TX gui lenh sang GSM RX
#use rs232(baud=9600, xmit=PIN_C6, rcv=PIN_C7, stream=COM, ERRORS)

//================ PIN DEFINE =================
#define LCD_ADDR       0x4E

#define STOP_BUTTON    PIN_B4
#define BUZZER         PIN_B5

#define ENGINE_BUTTON  PIN_D0   // Nut nhan bat/tat may
#define LED_GREEN      PIN_D1   // Den xanh bao dong co dang bat
#define GSM_PWRKEY     PIN_D3   // PWRKEY GSM A7680C

#define SAFE_RADIUS_M  10.0     // Ban kinh an toan 10 met
#define OUT_CONFIRM    3        // Vuot 10m lien tiep 3 lan moi gui SMS

char PHONE_NUMBER[] = "+84906425870";

//================ GPS VARIABLES =================
char gps_lat[16];
char gps_lon[16];
char gps_ns = 'N';
char gps_ew = 'E';

char base_lat_raw[16];
char base_lon_raw[16];
char base_ns = 'N';
char base_ew = 'E';

char last_type[6];
char last_rmc_status = '?';
char last_gga_fix = '?';

int1 gps_seen_data = 0;
int1 gps_has_position = 0;
int1 origin_saved = 0;

float current_lat_dec = 0;
float current_lon_dec = 0;
float base_lat_dec = 0;
```

```
52          float base_lon_dec = 0;
53
54          //================ LCD =================
55          void lcd_i2c_write(int8 data) {
56              i2c_start();
57              i2c_write(LCD_ADDR);
58              i2c_write(data);
59              i2c_stop();
60          }
61
62          void lcd_enable(int8 data) {
63              lcd_i2c_write(data | 0x04);
64              delay_us(2);
65              lcd_i2c_write(data & ~0x04);
66              delay_us(50);
67          }
68
69          void lcd_send4(int8 nibble, int1 rs) {
70              int8 data = 0;
71
72              if(rs) data |= 0x01;
73              data |= 0x08;
74
75              if(nibble & 0x01) data |= 0x10;
76              if(nibble & 0x02) data |= 0x20;
77              if(nibble & 0x04) data |= 0x40;
78              if(nibble & 0x08) data |= 0x80;
79
80              lcd_enable(data);
81          }
82
83          void lcd_send8(int8 value, int1 rs) {
84              lcd_send4(value >> 4, rs);
85              lcd_send4(value & 0x0F, rs);
86          }
87
88          void lcd_cmd(int8 cmd) {
89              lcd_send8(cmd, 0);
90              delay_ms(2);
91          }
92
93          void lcd_putc(char c) {
94              lcd_send8(c, 1);
95          }
96
97          void lcd_clear() {
98              lcd_cmd(0x01);
99              delay_ms(2);
100         }
101
102         void lcd_gotoxy(int8 x, int8 y) {
103             if(y == 1) {
104                 lcd_cmd(0x80 + x - 1);
105             }
106             else {
```

```
107                 lcd_cmd(0xC0 + x - 1);
108             }
109         }
110 
111         void lcd_init() {
112             delay_ms(50);
113 
114             lcd_send4(0x03, 0);
115             delay_ms(5);
116             lcd_send4(0x03, 0);
117             delay_ms(5);
118             lcd_send4(0x03, 0);
119             delay_ms(5);
120             lcd_send4(0x02, 0);
121             delay_ms(5);
122 
123             lcd_cmd(0x28);
124             lcd_cmd(0x0C);
125             lcd_cmd(0x06);
126             lcd_clear();
127         }
128 
129         //================ STRING HELPERS =================
130         void uart_write_ram(char *s) {
131             int8 i = 0;
132 
133             while(s[i] != 0) {
134                 fputc(s[i], COM);
135                 i++;
136             }
137         }
138 
139         void copy_text(char *src, char *dest, int8 max_len) {
140             int8 i = 0;
141 
142             while(src[i] != 0 && i < (max_len - 1)) {
143                 dest[i] = src[i];
144                 i++;
145             }
146 
147             dest[i] = 0;
148         }
149 
150         //================ GPS =================
151         void clear_gps_data() {
152             int8 i;
153 
154             for(i = 0; i < 16; i++) {
155                 gps_lat[i] = 0;
156                 gps_lon[i] = 0;
157                 base_lat_raw[i] = 0;
158                 base_lon_raw[i] = 0;
159             }
160 
161             gps_ns = 'N';
```

```
162             gps_ew = 'E';
163             base_ns = 'N';
164             base_ew = 'E';
165         }
166 
167         void clear_type() {
168             int8 i;
169 
170             for(i = 0; i < 6; i++) {
171                 last_type[i] = 0;
172             }
173         }
174 
175         float str_to_float(char *s) {
176             int8 i = 0;
177             int1 after_dot = 0;
178             float value = 0;
179             float div = 10.0;
180 
181             while(s[i] != 0) {
182                 if(s[i] == '.') {
183                     after_dot = 1;
184                 }
185                 else if(s[i] >= '0' && s[i] <= '9') {
186                     if(!after_dot) {
187                         value = value * 10.0 + (s[i] - '0');
188                     }
189                     else {
190                         value = value + ((float)(s[i] - '0') / div);
191                         div = div * 10.0;
192                     }
193                 }
194 
195                 i++;
196             }
197 
198             return value;
199         }
200 
201         float nmea_to_decimal(char *raw, char dir) {
202             float v;
203             float minutes;
204             float decimal;
205             int16 degrees;
206 
207             if(raw[0] == 0) {
208                 return 0;
209             }
210 
211             v = str_to_float(raw);
212 
213             degrees = (int16)(v / 100.0);
214             minutes = v - ((float)degrees * 100.0);
215 
216             decimal = (float)degrees + (minutes / 60.0);
```

```
217
218             if(dir == 'S' || dir == 'W') {
219                 decimal = 0 - decimal;
220             }
221
222             return decimal;
223         }
224
225         int1 update_decimal_position() {
226             if(gps_lat[0] == 0 || gps_lon[0] == 0) {
227                 return 0;
228             }
229
230             current_lat_dec = nmea_to_decimal(gps_lat, gps_ns);
231             current_lon_dec = nmea_to_decimal(gps_lon, gps_ew);
232
233             gps_has_position = 1;
234             return 1;
235         }
236
237         float distance_meter(float lat1, float lon1, float lat2, float lon2) {
238             float dlat_m;
239             float dlon_m;
240             float mid_lat;
241             float dis;
242
243             mid_lat = (lat1 + lat2) / 2.0;
244
245             dlat_m = (lat2 - lat1) * 111320.0;
246             dlon_m = (lon2 - lon1) * 111320.0 * cos(mid_lat * 0.0174532925);
247
248             dis = sqrt((dlat_m * dlat_m) + (dlon_m * dlon_m));
249
250             return dis;
251         }
252
253         void copy_field(char *line, int8 target_field, char *out, int8 max_len) {
254             int8 i = 0;
255             int8 field = 0;
256             int8 pos = 0;
257             char c;
258
259             out[0] = 0;
260
261             while(line[i] != 0) {
262                 c = line[i];
263
264                 if(c == ',') {
265                     if(field == target_field) {
266                         out[pos] = 0;
267                         return;
268                     }
269
270                     field++;
271                     pos = 0;
```

```
                i++;
                continue;
            }

            if(c == '*' || c == '\r' || c == '\n') {
                if(field == target_field) {
                    out[pos] = 0;
                    return;
                }

                return;
            }

            if(field == target_field) {
                if(pos < (max_len - 1)) {
                    out[pos] = c;
                    pos++;
                    out[pos] = 0;
                }
            }

            i++;
        }

        out[pos] = 0;
    }

    int1 gps_read_line(char *line, int8 max_len, unsigned int16 timeout_ms) {
        unsigned int16 t = 0;
        int8 idx = 0;
        int1 started = 0;
        char c;

        while(t < timeout_ms) {
            if(kbhit(COM)) {
                c = fgetc(COM);
                gps_seen_data = 1;

                if(c == '$') {
                    started = 1;
                    idx = 0;
                    line[idx++] = c;
                }
                else if(started) {
                    if(idx < (max_len - 1)) {
                        line[idx++] = c;
                    }

                    if(c == '\r' || c == '\n') {
                        line[idx] = 0;
                        return 1;
                    }
                }
            }

```

```
327                 delay_ms(1);
328                 t++;
329             }
330 
331             line[0] = 0;
332             return 0;
333         }
334 
335         int1 is_rmc(char *line) {
336             if(line[0] == '$' && line[3] == 'R' && line[4] == 'M' && line[5] == 'C') {
337                 return 1;
338             }
339 
340             return 0;
341         }
342 
343         int1 is_gga(char *line) {
344             if(line[0] == '$' && line[3] == 'G' && line[4] == 'G' && line[5] == 'A') {
345                 return 1;
346             }
347 
348             return 0;
349         }
350 
351         void save_type(char *line) {
352             int8 i;
353 
354             clear_type();
355 
356             for(i = 0; i < 5; i++) {
357                 last_type[i] = line[i + 1];
358             }
359 
360             last_type[5] = 0;
361         }
362 
363         // RMC:
364         // field 0 = $GNRMC
365         // field 1 = time
366         // field 2 = status A/V
367         // field 3 = lat
368         // field 4 = N/S
369         // field 5 = lon
370         // field 6 = E/W
371         int1 parse_rmc(char *line) {
372             char status[3];
373             char ns[3];
374             char ew[3];
375 
376             save_type(line);
377 
378             copy_field(line, 2, status, 3);
379             last_rmc_status = status[0];
380 
381             if(status[0] != 'A') {
```

```
382             return 0;
383         }
384 
385         copy_field(line, 3, gps_lat, 16);
386         copy_field(line, 4, ns, 3);
387         copy_field(line, 5, gps_lon, 16);
388         copy_field(line, 6, ew, 3);
389 
390         gps_ns = ns[0];
391         gps_ew = ew[0];
392 
393         return update_decimal_position();
394     }
395 
396     // GGA:
397     // field 0 = $GNGGA
398     // field 1 = time
399     // field 2 = lat
400     // field 3 = N/S
401     // field 4 = lon
402     // field 5 = E/W
403     // field 6 = fix quality
404     int1 parse_gga(char *line) {
405         char fix[3];
406         char ns[3];
407         char ew[3];
408 
409         save_type(line);
410 
411         copy_field(line, 6, fix, 3);
412         last_gga_fix = fix[0];
413 
414         if(fix[0] == '0' || fix[0] == 0) {
415             return 0;
416         }
417 
418         copy_field(line, 2, gps_lat, 16);
419         copy_field(line, 3, ns, 3);
420         copy_field(line, 4, gps_lon, 16);
421         copy_field(line, 5, ew, 3);
422 
423         gps_ns = ns[0];
424         gps_ew = ew[0];
425 
426         return update_decimal_position();
427     }
428 
429     int1 gps_get_position(unsigned int16 timeout_ms) {
430         char line[90];
431 
432         if(gps_read_line(line, 90, timeout_ms)) {
433             if(is_rmc(line)) {
434                 if(parse_rmc(line)) {
435                     return 1;
436                 }
```

```
437                 }
438
439                 if(is_gga(line)) {
440                     if(parse_gga(line)) {
441                         return 1;
442                     }
443                 }
444             }
445
446             return 0;
447         }
448
449         int1 wait_gps_position(unsigned int16 seconds) {
450             unsigned int16 sec;
451
452             for(sec = 0; sec < seconds; sec++) {
453                 lcd_clear();
454
455                 if(gps_seen_data) {
456                     lcd_gotoxy(1,1);
457                     printf(lcd_putc, "TYPE:%s", last_type);
458                     lcd_gotoxy(1,2);
459                     printf(lcd_putc, "R:%c G:%c S:%Lu", last_rmc_status, last_gga_fix,
                            sec);
460                 }
461                 else {
462                     lcd_gotoxy(1,1);
463                     printf(lcd_putc, "NO GPS DATA");
464                     lcd_gotoxy(1,2);
465                     printf(lcd_putc, "GPS TX->RC7");
466                 }
467
468                 if(gps_get_position(1000)) {
469                     return 1;
470                 }
471             }
472
473             return 0;
474         }
475
476         //================ LCD SCREEN =================
477         void show_engine_gps() {
478             lcd_clear();
479
480             if(gps_has_position) {
481                 lcd_gotoxy(1,1);
482                 printf(lcd_putc, "LA:%s%c", gps_lat, gps_ns);
483                 lcd_gotoxy(1,2);
484                 printf(lcd_putc, "LO:%s%c", gps_lon, gps_ew);
485             }
486             else {
487                 lcd_gotoxy(1,1);
488                 printf(lcd_putc, "DONG CO BAT");
489                 lcd_gotoxy(1,2);
490                 printf(lcd_putc, "DANG LAY GPS");
```

```
491             }
492         }
493
494         void show_origin_saved() {
495             lcd_clear();
496             lcd_gotoxy(1,1);
497             printf(lcd_putc, "DA LUU TOA DO");
498             lcd_gotoxy(1,2);
499             printf(lcd_putc, "GOC AN TOAN");
500         }
501
502         void show_wait_origin() {
503             lcd_clear();
504             lcd_gotoxy(1,1);
505             printf(lcd_putc, "XE TAT MAY");
506             lcd_gotoxy(1,2);
507             printf(lcd_putc, "LAY GPS GOC");
508         }
509
510         void show_armed_distance(unsigned int16 d) {
511             lcd_clear();
512             lcd_gotoxy(1,1);
513             printf(lcd_putc, "CHONG TROM BAT");
514             lcd_gotoxy(1,2);
515             printf(lcd_putc, "D:%Lum R:10m", d);
516         }
517
518         void show_alarm_screen() {
519             lcd_clear();
520             lcd_gotoxy(1,1);
521             printf(lcd_putc, "VUOT BAN KINH");
522             lcd_gotoxy(1,2);
523             printf(lcd_putc, "GUI SMS CANH BAO");
524         }
525
526         void show_sms_done() {
527             lcd_clear();
528             lcd_gotoxy(1,1);
529             printf(lcd_putc, "DA GUI SMS");
530             lcd_gotoxy(1,2);
531             printf(lcd_putc, "CHECK PHONE");
532         }
533
534         //================ INPUT =================
535         int1 is_engine_button_pressed() {
536             return input(ENGINE_BUTTON);
537         }
538
539         int1 is_stop_pressed() {
540             return input(STOP_BUTTON);
541         }
542
543         //================ BUZZER =================
544         void buzzer_beep_continuous() {
545             output_high(BUZZER);
```

```
546             delay_us(800);
547             output_low(BUZZER);
548             delay_us(800);
549         }
550 
551         //================ GSM POWER =================
552         void gsm_power_on() {
553             lcd_clear();
554             lcd_gotoxy(1,1);
555             printf(lcd_putc, "BAT GSM...");
556             lcd_gotoxy(1,2);
557             printf(lcd_putc, "PWR HIGH");
558 
559             output_low(GSM_PWRKEY);
560             delay_ms(500);
561 
562             output_high(GSM_PWRKEY);
563             delay_ms(6000);
564 
565             output_low(GSM_PWRKEY);
566 
567             lcd_clear();
568             lcd_gotoxy(1,1);
569             printf(lcd_putc, "CHO GSM BOOT");
570             lcd_gotoxy(1,2);
571             printf(lcd_putc, "WAIT 20S");
572 
573             delay_ms(20000);
574         }
575 
576         //================ GSM SEND SMS =================
577         void send_sms_geofence() {
578             lcd_clear();
579             lcd_gotoxy(1,1);
580             printf(lcd_putc, "GUI SMS...");
581             lcd_gotoxy(1,2);
582             printf(lcd_putc, "CANH BAO GPS");
583 
584             fprintf(COM, "AT\r");
585             delay_ms(3000);
586 
587             fprintf(COM, "ATE0\r");
588             delay_ms(2000);
589 
590             fprintf(COM, "AT+CMGF=1\r");
591             delay_ms(3000);
592 
593             fprintf(COM, "AT+CSCS=\"GSM\"\r");
594             delay_ms(2000);
595 
596             fprintf(COM, "AT+CMGS=\"");
597             uart_write_ram(PHONE_NUMBER);
598             fprintf(COM, "\"\r");
599 
600             delay_ms(8000);
```

```
601
602             fprintf(COM, "CANH BAO CHONG TROM!\r");
603             fprintf(COM, "XE DA DI CHUYEN RA KHOI BAN KINH 10M.\r");
604
605             fprintf(COM, "TOA DO GOC:\r");
606             fprintf(COM, "LAT: ");
607             uart_write_ram(base_lat_raw);
608             fputc(base_ns, COM);
609             fprintf(COM, "\r");
610
611             fprintf(COM, "LON: ");
612             uart_write_ram(base_lon_raw);
613             fputc(base_ew, COM);
614             fprintf(COM, "\r");
615
616             fprintf(COM, "TOA DO HIEN TAI:\r");
617             fprintf(COM, "LAT: ");
618             uart_write_ram(gps_lat);
619             fputc(gps_ns, COM);
620             fprintf(COM, "\r");
621
622             fprintf(COM, "LON: ");
623             uart_write_ram(gps_lon);
624             fputc(gps_ew, COM);
625             fprintf(COM, "\r");
626
627             fputc(26, COM);   // Ctrl + Z
628
629             show_sms_done();
630
631             delay_ms(20000);
632         }
633
634         //================ SAVE ORIGIN =================
635         void save_origin_position() {
636             base_lat_dec = current_lat_dec;
637             base_lon_dec = current_lon_dec;
638
639             copy_text(gps_lat, base_lat_raw, 16);
640             copy_text(gps_lon, base_lon_raw, 16);
641
642             base_ns = gps_ns;
643             base_ew = gps_ew;
644
645             origin_saved = 1;
646         }
647
648         //================ MAIN =================
649         void main() {
650             int1 engine_on_state;
651             int1 alarm_latched;
652             int1 sms_sent;
653             int8 out_count;
654             int8 last_state;
655
```

```
656             float dis;
657             unsigned int16 dis_int;
658 
659             // RC7 = RX nhan GPS
660             // RC6 = TX gui GSM
661             // RC4 = SDA, RC3 = SCL
662             set_tris_c(0b10011000);
663 
664             // RB4 input: STOP
665             // RB5 output: BUZZER
666             set_tris_b(0b00010000);
667 
668             // RD0 input: nut nhan
669             // RD1 output: LED xanh
670             // RD3 output: GSM PWRKEY
671             set_tris_d(0b00000001);
672 
673             output_low(BUZZER);
674             output_low(LED_GREEN);
675             output_low(GSM_PWRKEY);
676 
677             clear_gps_data();
678 
679             lcd_init();
680 
681             lcd_clear();
682             lcd_gotoxy(1,1);
683             printf(lcd_putc, "GPS GEOFENCE");
684             lcd_gotoxy(1,2);
685             printf(lcd_putc, "BAN KINH 10M");
686 
687             delay_ms(2000);
688 
689             gsm_power_on();
690 
691             engine_on_state = 0;
692             alarm_latched = 0;
693             sms_sent = 0;
694             out_count = 0;
695             last_state = 255;
696 
697             while(TRUE) {
698 
699                 //================================================
700                 // XU LY NUT DO: NHAN 1 LAN BAT MAY, NHAN LAN NUA TAT MAY
701                 //================================================
702                 if(is_engine_button_pressed()) {
703                     delay_ms(30);
704 
705                     if(is_engine_button_pressed()) {
706                         engine_on_state = !engine_on_state;
707 
708                         // Moi lan doi trang thai thi reset hien thi
709                         last_state = 255;
710 
```

```
                        // Neu vua bat may
                        if(engine_on_state) {
                            output_high(LED_GREEN);
                            output_low(BUZZER);

                            alarm_latched = 0;
                            sms_sent = 0;
                            out_count = 0;
                            origin_saved = 0;
                        }

                        // Neu vua tat may
                        else {
                            output_low(LED_GREEN);
                            output_low(BUZZER);

                            alarm_latched = 0;
                            sms_sent = 0;
                            out_count = 0;
                            origin_saved = 0;
                        }

                        while(is_engine_button_pressed()) {
                            gps_get_position(100);
                            delay_ms(5);
                        }

                        delay_ms(30);
                    }
                }

                //================================================
                // TRANG THAI BAT MAY
                // Cap nhat GPS lien tuc, LED xanh sang
                //================================================
                if(engine_on_state) {
                    output_high(LED_GREEN);
                    output_low(BUZZER);

                    gps_get_position(1000);

                    if(last_state != 1) {
                        show_engine_gps();
                        last_state = 1;
                    }

                    if(gps_has_position) {
                        show_engine_gps();
                    }

                    delay_ms(500);
                }

                //================================================
                // TRANG THAI TAT MAY
```

```
766                 // Luu toa do goc, sau do kiem tra ban kinh 10m
767                 //================================================
768                 else {
769                     output_low(LED_GREEN);
770 
771                     // Neu dang bao dong, nhan STOP de tat coi
772                     if(alarm_latched && is_stop_pressed()) {
773                         delay_ms(20);
774 
775                         if(is_stop_pressed()) {
776                             alarm_latched = 0;
777                             output_low(BUZZER);
778 
779                             while(is_stop_pressed()) {
780                                 gps_get_position(100);
781                                 delay_ms(5);
782                             }
783 
784                             delay_ms(20);
785                             last_state = 255;
786                         }
787                     }
788 
789                     // Chua luu toa do goc thi lay GPS de luu
790                     if(!origin_saved) {
791                         if(last_state != 2) {
792                             show_wait_origin();
793                             last_state = 2;
794                         }
795 
796                         gps_get_position(1500);
797 
798                         if(gps_has_position) {
799                             save_origin_position();
800                             show_origin_saved();
801                             delay_ms(2000);
802                             last_state = 255;
803                         }
804                         else {
805                             // Neu chua co GPS thi tiep tuc cho
806                             wait_gps_position(5);
807 
808                             if(gps_has_position) {
809                                 save_origin_position();
810                                 show_origin_saved();
811                                 delay_ms(2000);
812                                 last_state = 255;
813                             }
814                         }
815 
816                         continue;
817                     }
818 
819                     // Da co toa do goc thi cap nhat toa do hien tai
820                     if(gps_get_position(1000)) {
```

```
821                         dis = distance_meter(base_lat_dec, base_lon_dec,
822                         current_lat_dec, current_lon_dec);
823
824                         dis_int = (unsigned int16)dis;
825
826                         show_armed_distance(dis_int);
827
828                         if(dis > SAFE_RADIUS_M) {
829                             out_count++;
830                         }
831                         else {
832                             out_count = 0;
833                         }
834
835                         // Vuot ban kinh 10m lien tiep vai lan moi canh bao
836                         if(out_count >= OUT_CONFIRM) {
837                             alarm_latched = 1;
838                         }
839                     }
840
841                     // Neu vuot ban kinh thi gui SMS 1 lan
842                     if(alarm_latched) {
843                         if(!sms_sent) {
844                             output_low(BUZZER);
845                             show_alarm_screen();
846
847                             send_sms_geofence();
848
849                             sms_sent = 1;
850                             last_state = 255;
851                         }
852
853                         // Sau khi gui SMS, coi tiep tuc keu cho den khi nhan STOP
854                         buzzer_beep_continuous();
855                     }
856                     else {
857                         output_low(BUZZER);
858                         delay_ms(300);
859                     }
860                 }
861             }
862         }
```